

Green energy transition and vulnerability to external shocks

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Abstract

We develop an endogenous growth model with imported dirty energy, domestically produced clean energy, and carbon taxes to study the macroeconomic implications of a green energy transition. Calibrated to Spain, the model quantifies how carbon taxes affect long-run outcomes and short-run resilience to external energy price shocks. While long-run gross domestic product (GDP) losses are modest and partly offset by endogenous gains in energy efficiency and clean investment, our key finding is that the transition delivers substantial short-run benefits. As the clean-energy share rises from 26% to 86%, exposure to international dirty-energy price shocks declines sharply, reducing GDP impacts from -0.43% to -0.08% . Carbon-tax-induced decoupling of clean from dirty-energy prices plays a central role in insulating the economy. Our results highlight an overlooked channel through which climate policy enhances macroeconomic stability.

Keywords: energy prices, green transition, external shocks, carbon tax

JEL Classification: O38, O52, O44, E32

1 Introduction

A primary objective of the green transition is to reduce long-term climate-related risks by lowering reliance on carbon-intensive energy while managing the short-run macroeconomic costs of adjustment. Accordingly, a large macroeconomic literature examines climate policies that support the transition (e.g., carbon taxes) primarily through their long-run and direct effects on allocations, growth, and welfare (e.g. [Acemoglu et al. 2012](#); [Golosov et al. 2014](#)). However, much less attention has been paid to how the green transition reshapes short-run macroeconomic fluctuations by altering the economy’s exposure to external energy price disturbances. This paper fills this gap by showing that climate policy affects not only long-run allocations but also business-cycle vulnerability in economies reliant on imported fossil fuels.

We study the ability of green transition to reduce the economy’s vulnerability to international dirty-energy price shocks in a calibrated endogenous growth model in which firms use imported dirty energy and domestically produced clean energy as production inputs. Our core exercise compares the transmission of an identical international dirty-energy price shock across three transition states that differ in the economy’s clean-energy share and are generated by different carbon-tax levels. Our central finding is that the green transition insulates the domestic economy from international dirty-energy price shocks: as the clean-energy share rises from 26% to 86%, the impact response of gross domestic product (GDP) falls from -0.43% to -0.08% for the same shock. A key mechanism is reduced pass-through from international dirty-energy prices to clean-energy prices (price decoupling), which dampens increases in firms’ average energy costs. Thus, climate policy affects not only long-run allocations but also short-run macroeconomic vulnerability to external energy price fluctuations.

More specifically, we consider a real model of firm dynamics based on [Atkeson and](#)

Burstein (2019) and Domínguez-Díaz et al. (2025), extended with energy inputs as in Airaudo et al. (2023). Firms use physical capital, labor, and energy as production inputs. Energy is sourced from imported dirty energy and domestically produced clean energy, which uses capital as an input. The price of dirty energy faced by domestic firms has two components: an exogenous international price and a per-unit carbon tax imposed by the domestic fiscal authority, designed to capture key features of the E.U. Emissions Trading Scheme (ETS). In addition, firms engage in two types of innovative investment (Romer, 1990; Aghion and Howitt, 1992): one improves the efficiency of non-energy inputs (capital and labor), and the other improves energy-use efficiency in production.

In our baseline, we calibrate the model to Spain in 2019, a small open economy reliant on imported dirty energy, characterized by a low clean-energy share and high exposure to imported energy. We use aggregate data from 2000 to 2019 and firm-level data from the Central Balance Sheet Data Office, maintained by Banco de España, to discipline the parameters governing firm innovation. The calibrated economy features a clean-energy share of 26% and an imported-dirty-energy-to-GDP ratio of 3.35%, consistent with Spanish data.

To generate alternative starting points for the simulations, we consider a carbon-tax-driven green transition. Specifically, we increase the carbon tax to construct two scenarios: one with a medium clean-energy share of 45% and another with a high share of 86%, consistent with Spanish targets for 2030 and 2050, respectively. In the long run (i.e., balanced growth path (BGP)), the macroeconomic effects of carbon taxes confirm the results of Airaudo et al. (2023). Higher carbon taxes foster clean-energy investment and improve energy efficiency by encouraging firms to substitute away from dirty energy and mitigate rising costs through more efficient energy use. However, when revenues are recycled via lump-sum transfers, these gains do not fully offset the higher after-tax cost of dirty energy, and GDP falls by 1.38% (3.5%) in the medium (high) clean-energy scenario. Coun-

terfactual exercises further show that both energy-efficiency investment and clean-energy investment are essential for limiting GDP losses and reducing dirty-energy use.

We then turn to the main contribution of the paper: assessing how the green transition reduces vulnerability to shocks in the international price of dirty energy. Specifically, we analyze the propagation of an exogenous 10% increase in the international price of dirty energy across the three scenarios with low, medium, and high clean-energy shares.

We find that shifting from imported dirty energy to domestically produced clean energy substantially reduces vulnerability to this external shock. GDP falls by 0.43% on impact in the 2019 calibration but declines by only 0.29% and 0.08% in the medium and high clean-energy scenarios, respectively. These reductions persist across all horizons: the present discounted value of the GDP decline falls by 30% (80%) in the medium (high) clean-energy economy relative to the baseline.

The green transition shields the economy from external energy price fluctuations through two channels. First, reduced reliance on imported dirty energy lowers direct exposure to price shocks. Second, clean and dirty-energy prices become decoupled, limiting increases in average energy costs. The first channel operates mechanically: lower dependence on imported energy reduces cost sensitivity to price increases, holding clean-energy prices constant. The second channel arises because the carbon tax constitutes a larger share of the domestic dirty-energy price, reducing the responsiveness of after-tax prices to international fluctuations and limiting pass-through to clean-energy prices.

Counterfactual exercises show that price decoupling is central to these results. When pass-through is higher, GDP, investment, consumption, and hours respond more similarly across scenarios. Intuitively, without decoupling, external shocks raise production costs for both energy types, leading to substantial declines in investment and labor demand and a more pronounced downturn.

The remainder of the paper is organized as follows. Section 2 relates our results to the

literature. Section 3 presents the model, and Section 4 describes the calibration. Section 5 reports the main results. Section 6 concludes.

2 Related Literature

This paper contributes to the literature on the macroeconomic effects of the green transition by addressing both long- and short-run economic consequences, as well as to the literature on the propagation of energy price shocks. In doing so, we bridge two key strands.

First, we contribute to the literature examining the long-run and transitional implications of green transition policies. Our contribution lies not only in analyzing these direct effects but also in emphasizing the role of the green transition in reducing vulnerability to external energy price shocks.

For example, [Golosov et al. \(2014\)](#) studied the optimal taxation of fossil fuels, deriving a formula for optimal carbon taxes in a general equilibrium setting where fossil fuels generate a negative externality. [Acemoglu et al. \(2012\)](#) analyzed environmental sustainability in a model with directed technological change and clean- and dirty-energy inputs, highlighting the need for policies that redirect innovation toward clean technologies. [Ernst et al. \(2023\)](#) and [Hinterlang \(2024\)](#) developed multi-country, multi-sector models to examine cross-country spillovers from increases in carbon taxes.

A recent set of papers focused on the inflationary consequences of carbon taxes and their implications for monetary policy. For instance, [Nakov and Thomas \(2023\)](#) derived the optimal monetary policy response to carbon taxes in a new Keynesian global model, showing that it should prioritize price stability. [Del Negro et al. \(2023\)](#) emphasized the role of relative price stickiness between dirty and clean-energy sectors in shaping inflation dynamics in a production network framework.

Closer to our approach, [Airaudo et al. \(2023\)](#) studied a new Keynesian small open economy model with endogenous growth and both clean- and dirty-energy inputs. That paper examined the effects of environmental policies, including carbon taxes, on GDP, energy efficiency, inflation, and welfare over both the long run and the transition path. While our long-run results align with their findings on dirty-energy taxation, our focus is on how the green transition insulates the economy from external shocks through the decoupling of clean- and dirty-energy prices. By calibrating the model to Spain and conducting simulations, we show that increasing the clean-energy share reduces sensitivity to international dirty-energy-price shocks, generating business-cycle benefits.

Second, we contribute to the literature on the macroeconomic consequences of energy price fluctuations. We complement existing research by offering a new perspective on how the green transition reshapes the transmission of such shocks. For example, [Bachmann et al. \(2022\)](#) studied the effects of a disruption in Russian energy imports on Germany, while [Pieroni \(2023\)](#) examined the distributional consequences of energy prices in heterogeneous-agent models. Similarly, [Auclert et al. \(2023\)](#) analyzed the propagation of energy price increases in open economies with household heterogeneity, and [Bayer et al. \(2023\)](#) investigated fiscal responses to energy crises. Our contribution is to show that the green transition—through increasing the share of clean energy and reducing reliance on imported dirty energy—fundamentally alters the transmission of energy price shocks.

3 Model

We build on and extend the endogenous growth model of [Atkeson and Burstein \(2019\)](#), also used in [Domínguez-Díaz et al. \(2025\)](#), by incorporating energy production inputs as in [Airaudo et al. \(2023\)](#). The framework is a real model of a closed economy, apart from imported dirty energy. Firms use labor, capital, and an energy good to produce

output. The energy good is a composite of imported dirty energy and domestically produced clean energy. Additionally, firms engage in innovative investment to improve the efficiency with which they use both energy and non-energy inputs ([Aghion and Howitt, 1992](#); [Romer, 1990](#)). We next describe the main components of the model and the problems faced by economic agents, relegating a detailed characterization of first-order conditions to Appendix [A](#).

3.1 Households

The economy is populated by an infinitely lived representative household with time-separable preferences over per capita consumption C_t/H_t and hours worked:

$$\max_{C_t, K_{t+1}, K_{t+1}^C, B_{t+1}, L_t} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t H_t \left(\log \left(\frac{C_t}{H_t} \right) - \kappa \chi_t \frac{\left(\frac{L_t}{H_t} \right)^{1+\varphi}}{1+\varphi} \right) \quad \text{s.t.}$$

$$C_t + I_t + I_t^C + B_{t+1} = (1 - \tau_t^L) W_t L_t + (1 - \tau_t^{\text{Corp}}) \left(r_t^k K_t + r_t^C K_t^C \right) \quad (1)$$

$$+ (1 + R_{t-1}) B_t + D_t - T_t,$$

$$K_{t+1} = (1 - \delta_K) K_t + I_t - \Xi^I(I_t, K_t), \quad (2)$$

$$K_{t+1}^C = (1 - \delta_K) K_t^C + I_t^C - \Xi^I(I_t^C, K_t^C), \quad (3)$$

where $\beta \in (0,1)$ and H_t denotes population, which grows at an exogenous rate, g_H . χ_t is an endogenous preference shifter, modeled following [Galí et al. \(2012\)](#), defined as,

$$\chi_t = \frac{\mathcal{W}_t}{C_t}, \quad (4)$$

$$\mathcal{W}_t = \mathcal{W}_{t-1}^{1-\nu} C_t^\nu. \quad (5)$$

This preference shifter dampens short-term wealth effects (e.g., [Coenen et al. 2024](#)),

preventing unrealistic labor supply responses to external supply shocks while preserving consistency with long-run balanced growth. In (5), when $\nu = 1$ preferences reduce to the King–Plosser–Rebelo specification (King et al., 1988), whereas as $\nu \rightarrow 0$, they converge to the Greenwood–Hercowitz–Huffman specification (Greenwood et al., 1988), which eliminates short-run wealth effects.

The household saves in two types of physical capital. First, non-clean capital K_t is rented to firms at rate r_t^k . Second, clean capital K_t^C , earning return r_t^C , is rented to domestic clean-energy producers. The household can also hold government bonds B_t , which yield the risk-free return, R_t . Labor income is earned by supplying hours L_t at wage W_t and is taxed at rate τ_t^L , while capital income is taxed at rate τ_t^{Corp} . The term, T_t , denotes lump-sum taxes, and aggregate firm profits D_t are rebated lump sum to the household.

The capital accumulation equations are given by (2) and (3), where δ_K is the depreciation rate, I_t and I_t^C denote investment in non-green and green capital, and $\Xi^I(I_t, K_t)$ and $\Xi^I(I_t^C, K_t^C)$ capture capital adjustment costs.

3.2 Production Sector

The supply side of the economy features multiple layers of production. A final good producer combines energy and non-energy inputs to produce a final good used for consumption and investment. These inputs are actually bundles of intermediate energy and non-energy goods. Intermediate non-energy goods are produced using physical capital and production labor, while intermediate energy producers combine imported dirty energy with domestically produced clean energy. Both types of intermediate producers engage in innovative investment. Finally, the economy includes energy and non-energy research good producers, both of which use research labor as an input.

3.2.1 Final Good Producer

A competitive final good producer combines differentiated energy and non-energy intermediate bundles, denoted by $Y_{E,t}$ and $Y_{X,t}$, respectively, using a constant elasticity of substitution (CES) production function to produce the final good Y_t :

$$\max_{Y_{X,t}, Y_{E,t}} Y_t - P_{X,t}Y_{X,t} - P_{E,t}Y_{E,t} \quad \text{s.t.} \quad Y_t = \left[\theta^{\frac{1}{\sigma}} Y_{X,t}^{\frac{\sigma-1}{\sigma}} + (1-\theta)^{\frac{1}{\sigma}} Y_{E,t}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (6)$$

with $\sigma > 0$. Here, $P_{X,t}$ and $P_{E,t}$ denote the price indices of non-energy and energy bundles. The bundles, $Y_{X,t}$ and $Y_{E,t}$, are defined as

$$Y_{X,t} = \left[\sum_{z_x} M_{x,t}(z_x) y_{x,t}(z_x)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \quad (7)$$

$$Y_{E,t} = \left[\sum_{z_e} M_{e,t}(z_e) y_{e,t}(z_e)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \quad (8)$$

with $\rho > 1$. The variable, $y_{x,t}(z_x)$, denotes the output of a non-energy intermediate producer with productivity index z_x and price $p_{x,t}(z_x)$, while $M_{x,t}(z_x)$ denotes the mass of such goods. The corresponding variables for energy intermediate goods, with price $p_{e,t}(z_e)$, are defined analogously.

3.2.2 Non-Energy Intermediate Good Producers

Non-energy intermediate firms produce differentiated goods y_x using production labor l^p and physical capital k . The production technology for an intermediate good at time t is summarized by its productivity index, z_x :

$$y_{x,t}(z_x) = z_x k_t(z_x)^{\alpha} l_t^p(z_x)^{1-\alpha}, \quad (9)$$

with $\alpha \in (0,1)$. As in [Atkeson and Burstein \(2019\)](#), z_x has a countable support with grid elements $z_{x,n}$, and the highest element represents the frontier technology for each good. Given flexible capital and labor at the firm level, firms choose inputs to maximize per-period variable profits:

$$\pi_{x,t}(z_x) \equiv p_{x,t}(z_x)y_{x,t}(z_x) - W_t l_t^p(z_x) - r_t^k k_t(z_x). \quad (10)$$

Non-energy intermediate producers can undertake innovative investment by purchasing research goods. This investment generates new products, as in [Romer \(1990\)](#), or improves the efficiency of existing ones.

Both incumbent firms (those producing at times t and $t - 1$) and entrants (those not producing at $t - 1$) can invest in innovation. Entrants can only create products new to the economy.¹ Incumbents can also improve the efficiency of their existing products. An exogenous fraction, δ_0 , of incumbent-produced goods exits each period. Details of the innovation process are provided in [Appendix A.2](#).

3.2.3 Energy Intermediate Good Producers

An energy intermediate producer with productivity index z_e combines imported dirty energy, $\mathcal{D}_t(z_e)$, and clean energy, $C_t(z_e)$, according to the following CES production function:

$$y_{e,t}(z_e) = z_e \left[\theta_e^{\frac{1}{\sigma_e}} ((1 - \Gamma_{\mathcal{D},t}) \mathcal{D}_t(z_e))^{\frac{\sigma_e-1}{\sigma_e}} + (1 - \theta_e)^{\frac{1}{\sigma_e}} C_t(z_e)^{\frac{\sigma_e-1}{\sigma_e}} \right]^{\frac{\sigma_e}{\sigma_e-1}}, \quad (11)$$

¹The model could also be adapted to include business stealing, as in standard quality ladder models ([Klette and Kortum, 2004](#)). See [Atkeson and Burstein \(2019\)](#).

with $\sigma_e > 0$. The term, $\Gamma_{\mathcal{D},t}$, captures an adjustment cost associated with the share of dirty energy in total energy use and is given by

$$\Gamma_{\mathcal{D},t} = \frac{\gamma \mathcal{D}}{2} \left(\frac{\mathcal{D}_t(z_e)/(\mathcal{D}_t(z_e) + C_t(z_e))}{\mathcal{D}/(\mathcal{D} + C)} - 1 \right)^2, \quad (12)$$

where $\mathcal{D}/(\mathcal{D} + C)$ denotes the steady-state share of dirty energy (e.g., [Coenen et al. 2018](#)). By limiting substitution between dirty and clean energy in the short run, these adjustment costs play a central role—discussed further below—in transmitting dirty-energy price changes to clean-energy prices and, ultimately, to total energy costs. More precisely, they serve two purposes: they help match the observed pass-through from dirty-energy prices to average energy prices, and they are essential for the counterfactual exercises in [Section 5.3](#).

Energy intermediate producers purchase clean inputs at price $P_{C,t}$. The price of dirty energy has two components. First, the international price, $P_{\mathcal{D},t}^*$, is exogenous to the domestic economy and follows an $AR(1)$ process in logs. Second, the government imposes a per-unit carbon tax on dirty-energy imports, $\tau_t^{\mathcal{D}}$. The post-tax price faced by domestic firms is therefore

$$P_{\mathcal{D},t} = P_{\mathcal{D},t}^* + \tau_t^{\mathcal{D}}. \quad (13)$$

Per-period variable profits are

$$\pi_{e,t}(z_e) \equiv p_{e,t}(z_e)y_{e,t}(z_e) - P_{C,t}C_t(z_e) - P_{\mathcal{D},t}\mathcal{D}_t(z_e). \quad (14)$$

As with non-energy intermediate producers, energy producers can undertake innovative investment. Products exit at the same exogenous rate, δ_0 . For simplicity, we restrict innovation to new entrants, modeled analogously to entry in the non-energy sector. Details of the innovation process are provided in [Appendix A.2](#).

3.2.4 Clean Energy Production

Clean energy is produced domestically by a representative firm, while dirty energy is imported. We assume that clean energy is produced using only physical capital, according to

$$C_t = \Omega_c \left(K_t^c \right)^{\alpha_e}, \quad (15)$$

where Ω_c is a scaling constant, and $\alpha_e \in (0, 1)$.

The representative firm chooses capital demand by solving

$$\max_{K_t^c} P_{c,t} C_t - r_t^c K_t^c \quad \text{s.t.} \quad C_t = \Omega_c \left(K_t^c \right)^{\alpha_e}. \quad (16)$$

3.2.5 Research Good Producers

Non-energy research good. A representative non-energy research producer hires labor $L_{x,t}^r$ to produce the research good used in the innovation of non-energy intermediates:

$$Y_{x,t}^r = A_{x,t}^r Z_{x,t}^{\phi-1} L_{x,t}^r. \quad (17)$$

Here, $A_{x,t}^r$ represents a stock of freely available knowledge, assumed to grow at exogenous rate $g_{A_{r,x}}$. The term, $Z_{x,t}^{\phi-1}$, with $\phi \leq 1$, follows [Jones \(2002\)](#) and captures intertemporal knowledge spillovers. Since $\phi \leq 1$, higher aggregate productivity $Z_{x,t}$ reduces research efficiency, consistent with the idea that “ideas are getting harder to find” ([Bloom et al., 2020](#)).

The research good is sold at price $P_{x,t}^r$ to innovating firms, and the producer solves

$$\max_{L_{x,t}^r} P_{x,t}^r Y_{x,t}^r - W_t L_{x,t}^r \quad \text{s.t.} \quad Y_{x,t}^r = A_{x,t}^r Z_{x,t}^{\phi-1} L_{x,t}^r. \quad (18)$$

Energy research good. The energy research good is produced analogously:

$$Y_{e,t}^r = A_{e,t}^r Z_{e,t}^{\phi-1} L_{e,t}^r \quad (19)$$

where $L_{e,t}^r$ denotes research labor. As before, $\phi \leq 1$ captures intertemporal knowledge spillovers.

The research good is priced at $P_{e,t}^r$, and the producer solves

$$\max_{L_{e,t}^r} P_{e,t}^r Y_{e,t}^r - W_t L_{e,t}^r \quad \text{s.t.} \quad Y_{e,t}^r = A_{e,t}^r Z_{e,t}^{\phi-1} L_{e,t}^r. \quad (20)$$

3.3 Government

The government is represented by a fiscal authority that raises revenue from corporate taxes, carbon taxes on imported dirty energy, labor income taxes, and lump-sum taxes. These revenues, together with government debt B_{t+1} , finance interest payments on public debt. The government budget constraint is

$$\begin{aligned} B_t(1 + R_{t-1}) = & B_{t+1} + \tau_t^{\text{Corp}} \left(\sum_{z_x} M_t(z_x) \pi_t(z_x) + \sum_{z_e} M_t(z_e) \pi_t(z_e) + r_t^K K_t + r_t^C K_t^C \right) \\ & + \tau_t^{\mathcal{D}} \mathcal{D}_t + \tau_t^L W_t L_t + T_t, \end{aligned} \quad (21)$$

where τ_t^{Corp} is the corporate tax rate applied to firm profits.

3.4 Equilibrium, Market Clearing, and Productivity Dynamics

A *competitive equilibrium* is a sequence of consumption and labor allocations, $\{C_t, L_{e,t}^r, L_{x,t}^r, L_t^p\}_t$, and private and innovative investment, $\{I_t, I_t^C, x_{x,t}^n, x_{x,t}^m, x_{x,t}^c, x_{e,t}^n\}_t$, such that, given prices, taxes, and sequences for the international dirty-energy price and carbon tax, $\{P_{\mathcal{D},t}^*, \tau_{\mathcal{D},t}\}_t$,

households and firms optimize and markets clear. The aggregate resource constraint is

$$C_t + I_t + I_t^c = Y_t - P_{\mathcal{D},t}^* \mathcal{D}_t - \Xi^I(I_t^c, K_t^c) - \Xi^I(I_t, K_t). \quad (22)$$

Hence, output is allocated to consumption, investment in non-energy capital, investment in clean-energy capital, imports of dirty energy, and adjustment costs. We define *GDP* as the right-hand side of (22) (i.e., gross output net of adjustment costs and imported dirty energy used as an intermediate input).

A BGP is a competitive equilibrium in which all variables grow at constant rates. Appendix A.4 provides the detrended system and stationary equilibrium conditions, while Appendix A.3 derives analytical expressions for productivity dynamics in the energy and non-energy sectors.

4 Calibration

We calibrate the model to the Spanish economy using data from 2000–2019, beginning shortly after the creation of the Euro Area and ending prior to COVID-19. One model period corresponds to one year. We rely on two primary data sources. First, we use aggregate data from the National Statistical Office of Spain (*Instituto Nacional de Estadística*, INE). Second, we use firm-level data from the Central Balance Sheet Data Office (*Central de Balances*), maintained by Banco de España, to discipline the parameters governing firm innovation.

4.1 Household Sector

We set the inverse Frisch elasticity, φ , equal to one, consistent with the range of estimates in Chetty et al. (2011). The discount factor, β , is chosen to target an annualized

steady-state interest rate of 4.5%. Population growth g_H is set to 0.6%, matching the average growth rate in Spain over the sample period. The parameter ν , governing short-run wealth effects in (5), is set to 0.02 following Galí et al. (2012). The labor disutility parameter, κ , is calibrated to normalize hours worked to one at the BGP.

We assume the following functional form for capital adjustment costs (Christiano et al., 2011):

$$\Xi^I(I_t, K_t) = \frac{\sigma_I}{2} \left(\frac{I_t}{K_t} - (\delta_K + \exp(g_Y) - 1) \right)^2 K_t, \quad (23)$$

$$\Xi^I(I_t^c, K_t^c) = \frac{\sigma_I}{2} \left(\frac{I_t^c}{K_t^c} - (\delta_K + \exp(g_Y) - 1) \right)^2 K_t^c, \quad (24)$$

where g_Y denotes the constant growth rate of output along the BGP. We set $\sigma_I = 17$, in line with Eberly et al. (2008), and choose the depreciation rate, δ_K , to match an aggregate investment-to-GDP ratio of 21%.

4.2 Government

We set the corporate tax rate to $\tau^{\text{Corp}} = 0.25$, consistent with the standard flat rate for Spanish corporations, and the labor income tax rate to $\tau^L = 0.13$ to match the effective rate in Spain in 2019, as reported by the Spanish Tax Agency.²

We set the carbon tax to zero in the 2019 calibration. As discussed below, this instrument is used to target future clean-energy shares in the economy. Given these fiscal parameters, we normalize public debt to zero and set lump-sum taxes T_t to satisfy the government budget constraint at the BGP.

²See the Tax Revenue Annual Report of the 2019 Fiscal Year (https://sede.agenciatributaria.gob.es/Sede/en_gb/estadisticas/recaudacion-tributaria/informe-anual/ejercicio-2019.html) prepared by the Spanish Tax Agency (*Agencia Tributaria*).

4.3 Production and Innovation

Standard parameters. We set the elasticity of substitution across intermediate goods to four, in line with [Broda and Weinstein \(2006\)](#). We set α to match a labor share of 0.57, and the markup. μ . to target a capital-to-output ratio of 4.2, as in [Arencibia Pareja et al. \(2018\)](#).

We set the exogenous growth rate of $A_{r,x}$ to 0.01 to match an annual output growth rate of 1.6% at the BGP, consistent with the sample average. Appendix [A](#) shows that, at the BGP, output in the non-energy and energy sectors, Y_x and Y_e , must grow at the same rate. Accordingly, we calibrate the exogenous productivity growth rate of energy, $g_{Ar,e}$, to satisfy this condition.

Energy-related parameters. The quasi-share on non-energy goods in production, θ , is set to match an imported-energy-to-GDP ratio of 3.35% for Spain, following [Balteanu and Viani \(2023\)](#). The elasticity of substitution between energy and non-energy goods, σ , is set to 0.14, in line with the median estimates of [Labandeira et al. \(2017\)](#). We set the quasi-share of dirty-energy inputs, θ_e , to target a clean-energy share $\frac{c}{c+\mathcal{D}}$ of 26%, consistent with estimates from the Joint Research Center of the European Commission.³ The elasticity of substitution between dirty and clean energy is set to three, following [Papageorgiou et al. \(2017\)](#). The exponent of clean-energy capital in (15), α_e , is set to 0.9 to match the capital share of the energy sector in Organization for Economic Cooperation and Development (OECD) InterCountry Input–Output (ICIO) tables.

The adjustment cost parameter, $\gamma_{\mathcal{D}}$, in (12) reduces substitutability between dirty and clean energy, thereby limiting the pass-through of international dirty-energy prices to clean energy prices. This is a key mechanism, as firms use both energy inputs rather than relying solely on dirty energy.

³As explained later, the scenarios we consider are based on projections from the Joint Research Center of the European Commission. To maintain consistency, we use the same source for 2019. Our main results are not sensitive to these exact values.

According to [Quintana \(2024\)](#), clean-energy production reduced the average energy price by about 15% in Spain in 2019. We therefore calibrate $\gamma_{\mathcal{D}}$ such that, in response to a shock, the model-implied increase in the average energy price is 15% smaller than the increase in the international dirty-energy price.

Finally, we normalize clean and dirty-energy prices to one, so that $P_c = P_{\mathcal{D}}^* = 1$. The scaling parameter, Ω_c , is chosen to be consistent with this normalization.

Innovation-related parameters. In the research production function, ϕ governs the intertemporal externality of technological progress. Empirical evidence suggests that ϕ is negative (e.g., [Bloom et al. 2020](#)), implying that higher productivity reduces the efficiency of innovation, consistent with the notion that “ideas are getting harder to find.” Following [Fernald and Jones \(2014\)](#) and [Atkeson and Burstein \(2019\)](#), we set $\phi = -1.6$.⁴

The remaining innovation parameters, reported in the final panel of Table 1, follow the strategy of [Domínguez-Díaz et al. \(2025\)](#), which closely adheres to [Atkeson and Burstein \(2019\)](#). These parameters are calibrated to match firm-level moments from Spanish data. A detailed description of the procedure is provided in Appendix A.5.

⁴[Bloom et al. \(2020\)](#) reported a range of estimates across sectors and firms, with an aggregate estimate close to -2 (see [Jones 2022](#)), which is consistent with our calibration.

Parameter	Description	Value	Target / Source
Households			
φ	Frisch elasticity	1	Chetty et al. (2011)
β	Discount factor	0.97	$R = 4.5\%$
g_H	Growth rate Pop.	0.6%	INE
δ_K	Depreciation capital	3.24%	Invest.-GDP = 21%
σ_I	Capital adj. cost	17	Eberly et al. (2008)
ν	Labor supply pref. shifter	0.02	Galí et al. (2012)
Government			
τ^{Corp}	Corporate profit tax rate	0.25	Corp. Tax Rate Spain
τ^L	Labor income tax rate	0.13	Labor Inc. Tax Spain
$\tau^{\mathcal{D}}$	Carbon tax	0	
Production			
ρ	Elasticity Substitution	4	Broda and Weinstein (2006)
ϕ	Intertemp. Externality	-1.6	Fernald and Jones (2014)
α	Capital exponent prod. fct.	0.42	Labor Share = 57%
μ	Markup	1.07	Capital-GDP = 4.2
α_e	Capital share clean energy	0.90	ICIO tables
θ	Quasi-share non-energy	0.95	Imp. energy to GDP = 3.35%
θ_e	Quasi-share non-energy	0.74	Clean-to-dirty = 26%
σ	Elast. subst. $Y_x - Y_e$	0.14	Labandeira et al. (2017)
σ_e	Elast. subst. $C - \mathcal{D}$	3	Papageorgiou et al. (2017)
Ω_C	Scaling energy prod. fct.	0.09	$P_{\mathcal{D}} = P_C$
$g_{Ar,x}$	Non-energy exog. Prod. Growth	0.01	$g_Y = 1.6\%$
$g_{Ar,e}$	Energy exog. Prod. Growth	-0.002	$g_{Y_e} = g_{Y_x}$
Innovation			
δ_0	Exit rate	0.05	
$\eta_{n,x}, \eta_{n,e}$	Prod. step entrant	1.63	
$\eta_{m,x}$	Prod. step incumb.	0.75	See text for a discussion
$\{h_0, h_1\}$	Fct. innov. new prod. incumb.	$\{0.3, 0.4\}$	
$\{\zeta_0, \zeta_1, \zeta_2\}$	Fct. innov. exist. prod. incumb.	$\{0.97, 0.32, 0.5\}$	

Notes: List of calibrated parameters. See the main text for a discussion on targets, values, and data used.

Table 1: Calibration

5 Results

The main goal of this paper is to examine how the green transition reduces the economy's vulnerability to shocks in the international price of dirty energy. Specifically, we study how a given increase in the price of dirty energy propagates depending on the economy's long-run energy mix. We proceed in two steps. First, in Section 5.1, we construct three economies with different long-run energy mixes, achieved through varying carbon-tax levels. Second, in Section 5.2, we analyze how shocks to the international price of dirty energy propagate across these economies, which differ in their clean-energy shares.

5.1 Carbon Taxes and Green Transition

We consider a green transition driven by a carbon tax, a primary mitigation policy in Europe. In the baseline calibration, the Spanish economy in 2019 features a relatively low clean-energy share ($c/c+d = 26\%$) and a carbon tax τ^D normalized to zero. We then increase the carbon tax to generate two economies with higher clean-energy shares along their BGPs. The first features a medium clean-energy share of 45%, consistent with Spain's 2030 target, achieved with $\tau^D = 0.4$. The second features a high clean-energy share of 86%, consistent with the 2050 target, requiring $\tau^D = 2$.^{5 6}

To interpret the implied magnitudes, we map these carbon taxes into ETS-equivalent prices. In the 2019 calibration, imported energy inputs amount to € 43,880M, or 3.35% of GDP. Applying $\tau = 0.4$ yields revenues of € 17,552M, while $\tau = 2$ yields € 87,760M. Given that Spain emitted 101,663,873 t of CO₂ under the ETS in 2019, these revenues

⁵These targets were provided by the Joint Research Centre of the European Commission as part of a scenario analysis for Spain and are close to those in the Integrated Energy and Climate National Plan (PNIEC): https://commission.europa.eu/energy-climate-change-environment/implementation-eu-countries/energy-and-climate-governance-and-reporting/national-long-term-strategies_en.

⁶In the baseline, carbon-tax revenues are rebated lump sum to households. Appendix B reports results under an alternative specification in which labor income taxes are reduced.

	Clean Energy Share	Dirty energy over GDP	Total energy over GDP	GDP	Clean energy Price	Dirty energy Price (after tax)	Energy efficiency	Non-energy Efficiency
	(%, level)	(%, level)	(%, level)	(%, change)	(%, change)	(%, change)	(%, change)	(%, change)
Low clean energy Share ($\tau^D = 0$)	25.93	3.35	4.82	-	-	-	-	-
Medium clean energy Share ($\tau^D = 0.4$)	45.09	2.32	4.65	-1.38	5.37	40.00	6.28	-0.36
High clean energy Share ($\tau^D = 2$)	86.31	0.65	4.39	-3.50	14.46	200.00	17.22	-0.86

Notes: Aggregate effects of higher carbon taxes, calibrated to match different clean-energy shares as described in the main text. “Low clean energy share” corresponds to the 2019 baseline. “Medium clean energy share” and “High clean energy share” correspond to economies with clean-energy shares consistent with Spain’s 2030 and 2050 targets, respectively. The first three columns report levels at the BGP, while the remaining columns report percent deviations from the “Low clean energy share economy”.

Table 2: Long-run effects of higher carbon tax

correspond to ETS prices of 173 and 868 €/t, respectively. These values align with the net-zero scenario of the Network of Central Banks and Supervisors for Greening the Financial System (NGFS, 2024), which projects prices of around 200 \$/t by 2030 and above 700 \$/t by 2050.

Before analyzing shock transmission, we summarize in Table 2 the long-run macroeconomic effects of moving from the 2019 economy (low clean-energy share) to economies with medium and high clean-energy shares.

Overall, the long-run effects of higher carbon taxes confirm the findings of Airaud et al. (2023). Higher carbon taxes reduce dirty-energy use and stimulate clean-energy investment and energy efficiency. However, they also generate GDP losses in the long run due to higher after-tax energy prices.

More specifically, the first three columns of Table 2 capture the shift in the energy mix induced by the carbon tax, reporting the clean-energy share, the dirty-energy-to-GDP ratio, and the total energy-to-GDP ratio. As the carbon tax raises the effective price of dirty energy, the dirty-energy-to-GDP ratio declines sharply (Column 2). This decline is largely offset by increased domestic clean-energy production, resulting in only a modest reduction in total energy use relative to GDP (Column 3).

Higher energy prices increase production costs, reducing firms' investment in capital and labor (Columns 4–6). Firms partially offset these effects by increasing investment in energy efficiency (Column 7), at the expense of non-energy efficiency, highlighting the trade-offs involved in achieving a cleaner energy mix.

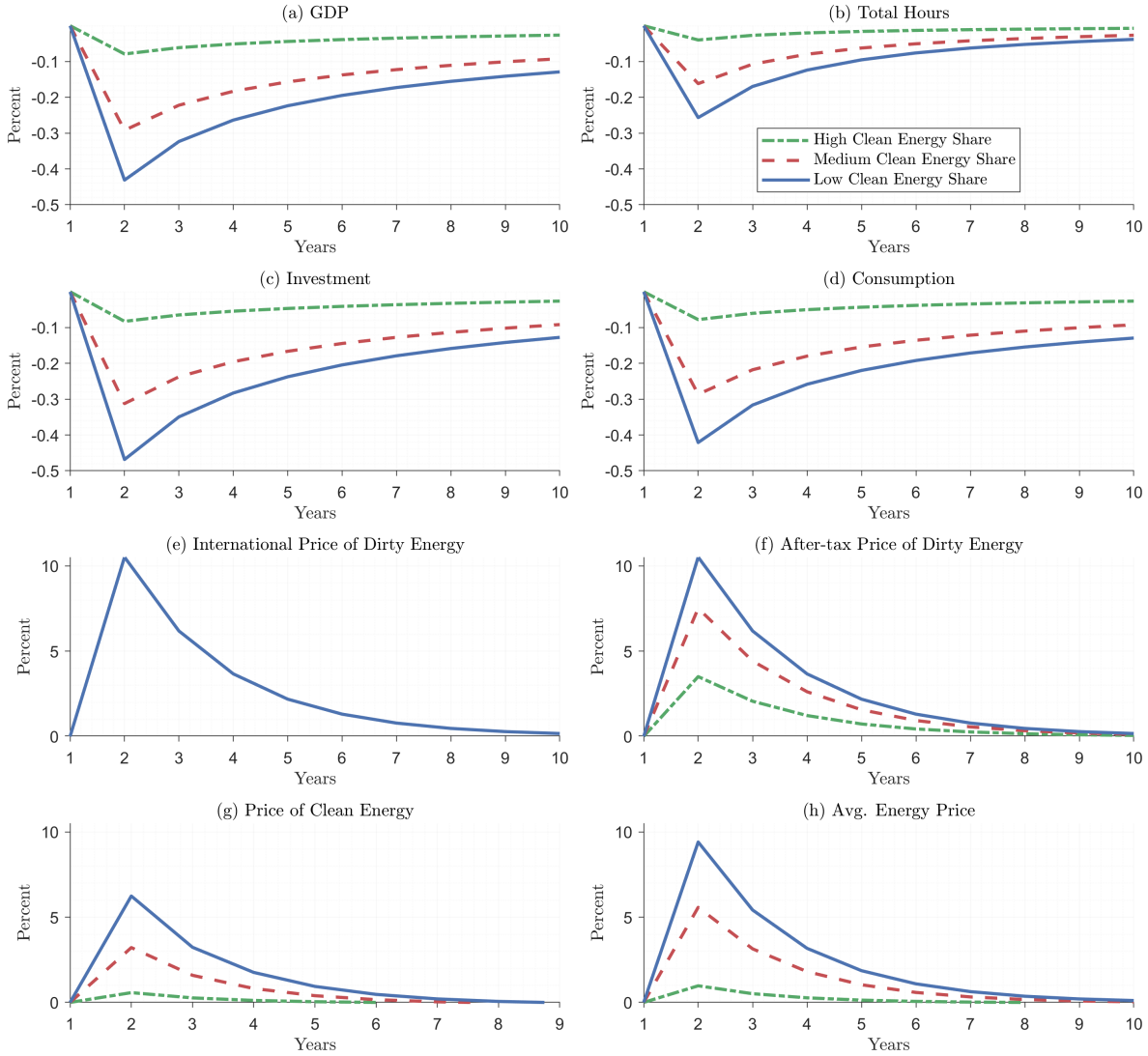
Appendix B quantifies the underlying mechanisms through counterfactual exercises. Eliminating endogenous investment in energy efficiency increases GDP losses by up to 60%. Similarly, shutting down clean-energy investment both amplifies GDP losses and limits the decline in dirty-energy use, as firms lose a substitute input. As a result, achieving the same reduction in dirty energy would require a higher carbon tax than in Table 2. Finally, recycling carbon-tax revenues through lower labor income taxes substantially mitigates GDP losses.

5.2 Green Transition and Vulnerability to External Shocks

Beyond long-run effects, the green transition also influences short-run macroeconomic fluctuations. In particular, a higher clean-energy share and reduced reliance on imported dirty energy can lower vulnerability to international energy price shocks. This section quantifies these gains and identifies the mechanisms driving them. Building on the long-run scenarios above, we examine how the short-run effects of an exogenous increase in the international price of dirty energy vary with the economy's energy mix.

Figure 1 shows the response to a 10% increase in the international price of dirty energy (Panel e), $P_{\mathcal{D},t}^*$, across the three scenarios. The blue solid line corresponds to the low clean-energy share economy, the dashed red line to the medium share economy, and the dotted green line to the high share economy, each evaluated at its respective BGP.

In the low clean-energy share economy, reflecting the Spanish baseline, a 10% increase in the international price of dirty energy reduces GDP by 0.43%. The shock operates through two channels. First, higher energy prices reduce households' real income, low-



Notes: Impulse response functions to a 10% increase in the international price of dirty energy (Panel e). Blue solid line: low clean-energy share economy. Red dashed line: medium clean share economy. Green dotted line: high clean share economy. Increases in the clean-energy share are achieved through higher carbon taxes; see Section 5 for details.

Figure 1: Effects of an increase in the international price of dirty energy

ering demand for domestically produced goods (Panel d). Second, increased production costs reduce firms' labor demand (Panel b) and investment (Panel c), leading to a decline in aggregate output. Due to capital adjustment costs, the short-run response is driven primarily by changes in hours worked, while medium-run dynamics reflect the persis-

tent decline in capital following lower investment.

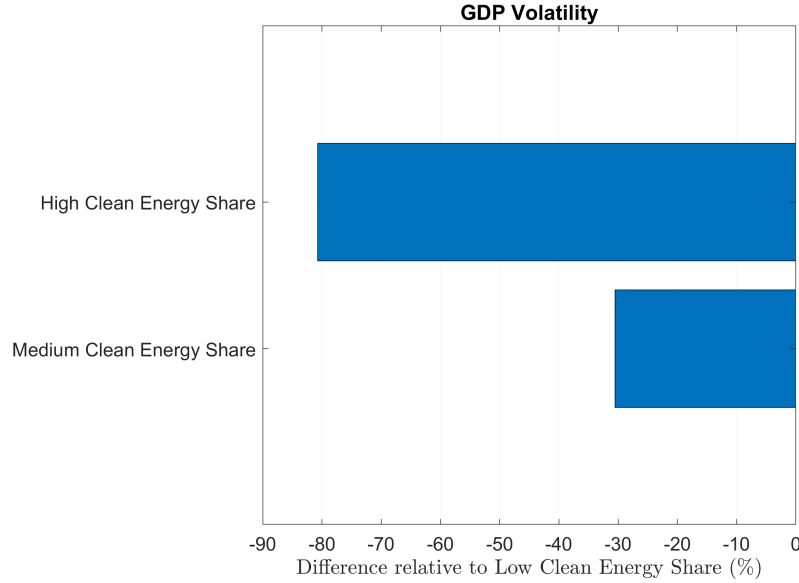
The magnitude of this effect is consistent with empirical evidence. For the U.S., [Känzig \(2021\)](#) finds that a 10% increase in oil prices reduces GDP by about 0.5%. For the Euro Area, [Peersman and van Robays \(2014\)](#) reports a decline of approximately 0.31%, with an estimate of 0.49% for Spain, closely matching our model. Our framework abstracts from nominal rigidities, which could amplify these effects. Although energy prices are relatively flexible ([Dhyne et al., 2006](#)), the decline in consumption and investment reduces aggregate demand. In a demand-driven setting, this could further depress output. However, as shown below, this demand effect weakens as the clean-energy share increases, suggesting that incorporating nominal rigidities would not alter our main conclusions.

The medium and high clean-energy scenarios show that the green transition substantially reduces vulnerability to energy price shocks. With a medium clean-energy share (aligned with 2030 targets), the same shock reduces GDP by 0.29%. With a high clean-energy share (aligned with 2050 targets), the effect is only 0.08%.

Two mechanisms drive these results. First, reliance on imported dirty energy declines sharply across scenarios (from 3.35% to 0.65% of GDP), reducing direct exposure to price shocks. Second, carbon taxes weaken the link between international dirty-energy prices and domestic clean-energy prices (Panel g). This reduced pass-through limits increases in the average energy price (Panel h), dampening production cost pressures and mitigating the downturn.

The green transition not only reduces the initial impact of shocks but also their persistence. The present discounted value of the GDP decline falls from -2.42% in the low clean share economy to -1.70% (a 30% reduction) and -0.48% (an 80% reduction) in the medium and high clean share economies, respectively.

These results are reinforced by the simulation in Fig. 2, which considers a sequence



Notes: Change in GDP volatility (measured by its standard deviation) in the medium and high clean-energy economies relative to the low clean-energy baseline. The standard deviation in each case is computed from simulations of a long series of shocks to the international price of dirty energy.

Figure 2: GDP volatility

of shocks to the international price of dirty energy.⁷ The contribution of these shocks to GDP volatility is 30% and 80% lower in the medium and high clean-energy economies, respectively, relative to the low clean-energy baseline.

Appendix C shows that the increased resilience to international dirty-energy price fluctuations as the clean-energy share rises is robust to several model extensions. In Appendix C.1, we incorporate trade in non-energy inputs and find that, although a richer trade structure alters the overall magnitude of the shock, the dampening effect of a higher clean-energy share remains. In Appendix C.2, we consider a clean-input subsidy as an alternative policy instrument. While this achieves the same energy mix as a carbon tax by promoting clean-energy production, albeit with higher fiscal costs, it yields similar short-run dynamics. Finally, Appendix C.3 shows that our results are robust to alterna-

⁷We simulate a series of normally distributed shocks to dirty-energy prices and apply the same sequence across all scenarios.

tive calibrations of the elasticity of substitution between clean and dirty energy, short-run adjustment costs, and innovation parameters.

5.3 Role of Reduced Dependence and Price Decoupling

The previous section identified two key mechanisms behind the reduced vulnerability to international dirty-energy price shocks following the green transition: lower reliance on imported dirty energy and the decoupling of clean- and dirty-energy prices, which limits increases in average energy costs.

Two factors prevent full pass-through from international dirty-energy prices to domestic clean-energy prices. First, once a carbon tax is introduced, a substantial portion of the domestic dirty-energy price ($P_{\mathcal{D},t}$) reflects the tax component ($\tau_t^{\mathcal{D}}$), reducing the relative importance of the pre-tax international price ($P_{\mathcal{D},t}^*$). As a result, the elasticity of the after-tax price with respect to international prices declines, limiting pass-through to clean-energy prices.⁸

Second, adjustment costs (see Eq. (12)) dampen the short-run substitution from dirty to clean energy following an increase in the price of the former. Combined with the inelastic short-run supply of clean energy, as it is produced solely with capital, this limits the increase in clean-energy prices.

In the baseline calibration, these adjustment costs are set to match empirical pass-through estimates (Quintana, 2024). In the counterfactual analysis, we set them to zero, thereby increasing the pass-through from international dirty-energy prices to clean-energy prices. This preserves the same energy mix across scenarios and isolates the role of price decoupling.

⁸Linearizing Eq. (13) around the BGP yields $\hat{p}_{\mathcal{D},t} = \frac{P_{\mathcal{D}}^*}{P_{\mathcal{D}}^* + \tau_{\mathcal{D}}} \hat{p}_{\mathcal{D},t}^*$, where hat variables denote deviations from the BGP. For a given $P_{\mathcal{D}}^*$, a higher $\tau_{\mathcal{D}}$ reduces the elasticity of the after-tax price with respect to the international price.

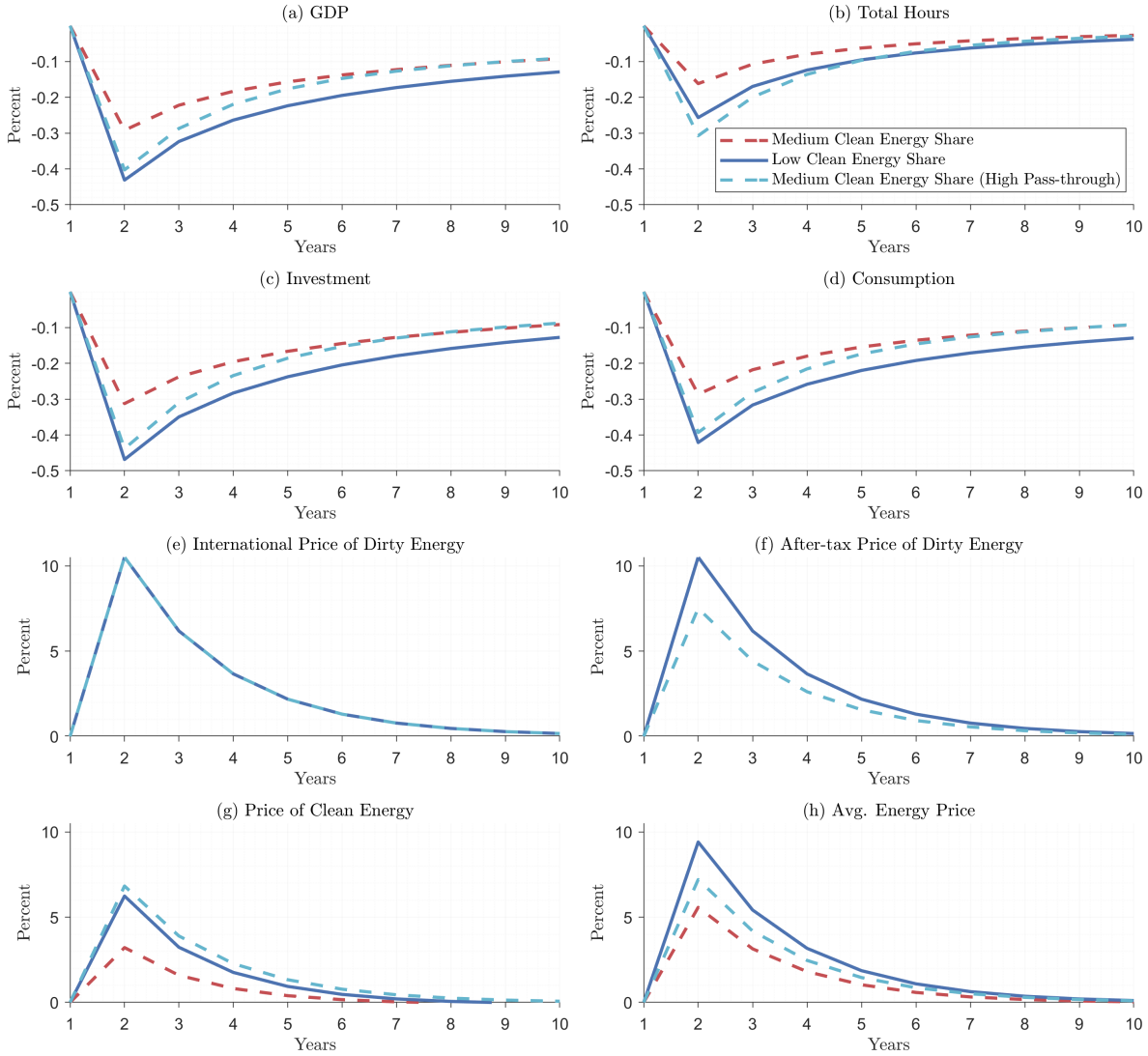


Figure 3: Effects of an increase in energy prices—The role of price decoupling

Figure 3 reports impulse responses for the low and medium clean-energy economies (solid blue and dashed red lines). The dashed light blue line shows the counterfactual with the same medium clean-energy share but no adjustment costs. In this case, price decoupling weakens, increasing the pass-through to clean-energy prices and, consequently, to the average energy price (compare dashed light blue and dashed red lines in Panels g and h). The response of the after-tax dirty-energy price is unchanged, as it is primarily determined by the carbon tax, which is identical across scenarios.

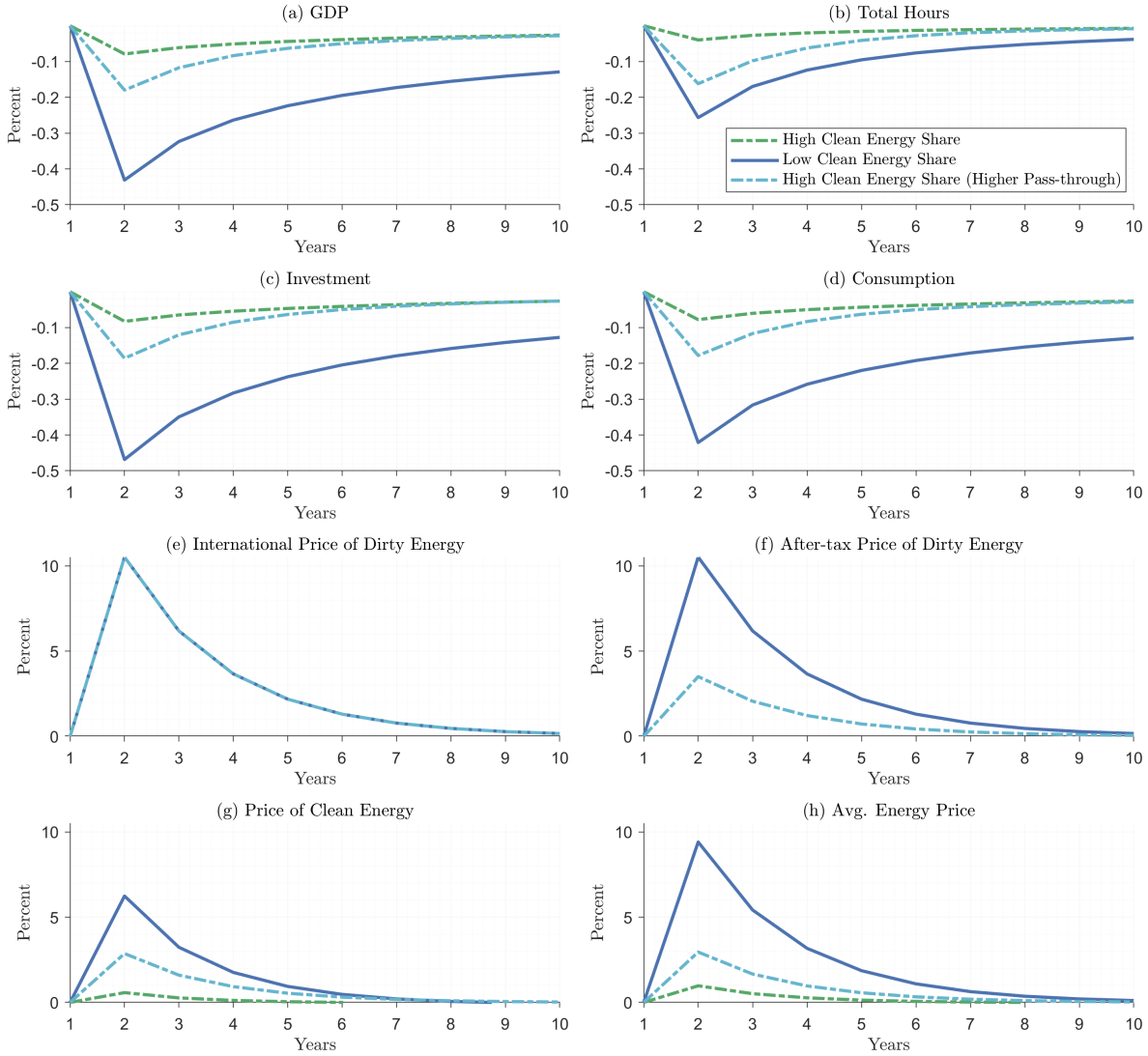


Figure 4: Effects of an increase in energy prices—The role of price decoupling

Price decoupling is central to the macroeconomic effects of the shock. When pass-through increases (Panel g), GDP, investment, consumption, and hours respond similarly to the low clean-energy case despite the higher clean-energy share (Panels a–d). Intuitively, even with reduced reliance on dirty energy, a shock that raises costs for both energy inputs significantly depresses investment and labor demand, resulting in a substantial downturn.

Figure 3 also shows that reduced dependence on dirty energy, alongside price decou-

pling, mitigates vulnerability to the shock. In the medium clean-energy scenario with high pass-through, GDP declines similarly on impact as in the low clean-energy case but recovers more quickly. This indicates that both channels—price decoupling and reduced dependence—jointly insulate the economy from fluctuations in the international price of dirty energy.

Figure 4 reinforces these findings by comparing the low clean-energy response (solid blue line) with the baseline high clean-energy response (dotted green line) and the high clean-energy response without adjustment costs (dotted blue line).

Even with a high clean-energy share, removing adjustment costs increases the response of clean-energy prices (Panel g). However, this increase remains smaller than in the low clean-energy case (see Fig. 3). This reflects the higher carbon tax in this scenario, which dampens the pass-through from international to after-tax dirty-energy prices (Panel f), and thus to clean-energy prices. In real terms, the higher pass-through significantly amplifies GDP losses following the energy price shock (Panel a).

Overall, the simulations show that the green transition yields substantial short-run benefits. Increasing the clean-energy share and reducing reliance on dirty energy lowers vulnerability to external energy price fluctuations. Counterfactual results further demonstrate that, in addition to reduced exposure to imported dirty energy, price decoupling is a key mechanism driving this reduction in vulnerability.

6 Conclusion

This paper examines how the green transition affects an economy’s vulnerability to external shocks in an endogenous growth model with domestically produced clean energy and imported dirty energy, calibrated to Spain. The results show that, although the transition to a greener energy mix entails modest long-run GDP costs, it delivers substantial bene-

fits by increasing resilience to external energy price volatility. In particular, higher carbon taxes that raise the clean-energy share and reduce reliance on dirty energy significantly mitigate GDP losses and stabilize the economy in response to fluctuations in international dirty-energy prices.

Our counterfactual analysis identifies two key mechanisms driving this reduction in vulnerability: lower direct dependence on imported dirty energy and the decoupling of dirty and clean-energy prices. Together, these channels dampen the transmission of international price shocks as the economy shifts toward cleaner energy. These findings suggest that policies reducing reliance on imported dirty energy and strengthening the decoupling of renewable energy prices from international fossil fuel prices can enhance macroeconomic stability. Even aside from climate considerations, such policies may be desirable as a means of reducing exposure to external shocks.

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A Model Appendix

In this appendix, we provide the detailed model specification outlined in the main text, including the first-order conditions associated with agents' optimization problems.

A.1 Households

The solution to the household problem in the main text satisfies the following optimality conditions:

$$\frac{1}{C_t/H_t} = \mathbb{E}_t \beta (1 + R_t) \frac{1}{C_{t+1}/H_{t+1}}, \quad (\text{A.1})$$

$$\frac{1}{C_t/H_t} = \mathbb{E}_t \beta \frac{1}{C_{t+1}/H_{t+1}} \frac{1}{q_t} \left(r_{t+1}^k (1 - \tau_{t+1}^{\text{Corp}}) + q_{t+1} \left(1 - \delta_K - \frac{\partial \Xi^I(I_{t+1}, K_{t+1})}{\partial K_{t+1}} \right) \right), \quad (\text{A.2})$$

$$\frac{1}{q_t} = 1 - \frac{\partial \Xi^I(I_t, K_t)}{\partial I_t}, \quad (\text{A.3})$$

$$\frac{1}{C_t/H_t} = \mathbb{E}_t \beta \frac{1}{C_{t+1}/H_{t+1}} \frac{1}{q_t^C} \left(r_{t+1}^C (1 - \tau_{t+1}^{\text{Corp}}) + q_{t+1}^C \left(1 - \delta_K - \frac{\partial \Xi^I(I_{t+1}^C, K_{t+1}^C)}{\partial K_{t+1}^C} \right) \right), \quad (\text{A.4})$$

$$\frac{1}{q_t^C} = 1 - \frac{\partial \Xi^I(I_t^C, K_t^C)}{\partial I_t^C}, \quad (\text{A.5})$$

$$\chi_t \kappa \left(\frac{L_t}{H_t} \right)^\varphi \frac{C_t}{H_t} = (1 - \tau_t^L) W_t, \quad (\text{A.6})$$

Equations (A.1), (A.2), and (A.4) are the Euler equations for risk-free bonds, physical capital, and clean capital, respectively. The variable, q_t , denotes Tobin's q —the marginal value of capital in consumption units—and is defined in Eq. (A.3). The analogous object for clean capital, q_t^C , is defined in equation (A.5). The final condition characterizes household labor supply.

A.2 Firms

A.2.1 Final Good Producer

The solution to the final good producer's problem, (6), yields the following system of demand equations for intermediate goods:

$$P_{x,t} = \left(\frac{Y_t}{Y_{x,t}} \right)^{\frac{1}{\sigma}} \theta^{\frac{1}{\sigma}}, \quad (\text{A.7})$$

$$P_{e,t} = \left(\frac{Y_t}{Y_{e,t}} \right)^{\frac{1}{\sigma}} (1 - \theta)^{\frac{1}{\sigma}}, \quad (\text{A.8})$$

$$y_{x,t}(z_x) = \left(\frac{p_{x,t}(z_x)}{P_{x,t}} \right)^{-\rho} Y_{x,t}, \quad (\text{A.9})$$

$$y_{e,t}(z_e) = \left(\frac{p_{e,t}(z_e)}{P_{e,t}} \right)^{-\rho} Y_{e,t}. \quad (\text{A.10})$$

A.2.2 Intermediate Good Producers

We decompose the problem of intermediate good producers into two parts. The first is a static problem in which firms choose production inputs and prices. The second is a dynamic problem in which firms decide on innovation investment.

Static problem: Non-energy intermediate good producers. A non-energy firm with productivity index z_x chooses labor $l^p(z_x)$, capital $k(z_x)$, and price $p_x(z_x)$ to maximize variable profits (10), subject to the demand system (A.7) and the production technology

(9). The solution is characterized by

$$\frac{k_t(z_x)}{l_t^p(z_x)} = \frac{\alpha}{1-\alpha} \frac{W_t}{r_t^k}, \quad (\text{A.11})$$

$$\frac{W_t^p}{p_{x,t}(z_x)} = \lambda_{x,t}(z_x)(1-\alpha) \frac{y_{x,t}(z_x)}{l_t^p(z_x)}, \quad (\text{A.12})$$

$$\lambda_{x,t}(z_x) = \max \left\{ 0, \frac{1}{z_x} \text{MC}_{x,t} \right\}, \quad (\text{A.13})$$

$$p_{x,t}(z_x) = \mu \frac{1}{z_x} \text{MC}_{x,t}, \quad (\text{A.14})$$

$$\mu = \min \left\{ \frac{\rho}{\rho-1}, \bar{\mu} \right\}, \quad (\text{A.15})$$

$$\text{MC}_{x,t} \equiv \left(\frac{W_t}{1-\alpha} \right)^{1-\alpha} \left(\frac{r_t^k}{\alpha} \right)^\alpha. \quad (\text{A.16})$$

Equation (A.11) shows that the capital-to-labor ratio chosen by intermediate firms is independent of the idiosyncratic productivity index, z_x , although the levels of these inputs may vary across firms. Equation (A.12) defines labor demand for a firm with productivity index z_x , where $\lambda_{x,t}(z_x)$ is the Lagrange multiplier given in (A.13).⁹

Equation (A.14) states that intermediate producers charge a constant markup μ over marginal cost $\text{MC}_{t/z}$, defined in (A.16). As in Atkeson and Burstein (2019), the markup is the minimum of the monopoly markup, $\rho/(\rho-1)$, and the technology gap relative to the second most productive firm producing the same good, with productivity index $z/\bar{\mu}$.

⁹The complementary slackness condition is $\lambda_{x,t}(z_x) \left(y_{x,t}(z_x) - z_x k_t(z_x)^\alpha l_t^p(z_x)^{1-\alpha} \right) = 0$.

Dynamic problem: Non-energy intermediate good producers. The dynamic problem of an incumbent non-energy firm consists of choosing innovative investment. We first describe the innovation process for entrants and incumbents, and then present the full dynamic problem.

Innovative investment by entering firms. Let $M_{x,t+1}^n$ denote the measure of non-energy entrants that invest in innovation at time t . Each entrant obtains at $t + 1$ a frontier technology to produce a new intermediate good with productivity index z'_x . As in [Atkeson and Burstein \(2019\)](#), and following [Luttmer \(2007\)](#), z'_x is drawn from a distribution such that $\mathbb{E}_t z_x'^{\rho-1} = \eta_n Z_{x,t}^{\rho-1} / M_{x,t}$, where $Z_{x,t}$ denotes aggregate productivity and $M_{x,t}$ the total measure of non-energy goods. Thus, $Z_{x,t} / M_{x,t}$ represents average productivity across existing intermediate goods. More precisely, these are defined as

$$Z_{x,t} = \left(\sum_{z_x} z_x^{\rho-1} M_{x,t}(z_x) \right)^{\frac{1}{\rho-1}}, \quad (\text{A.17})$$

$$M_{x,t} = \sum_{z_x} M_{x,t}(z_x). \quad (\text{A.18})$$

Finally, following [Atkeson and Burstein \(2019\)](#) and [Acemoglu \(2009\)](#), we assume that an entering firm must purchase $1/M_{x,t}$ units of the research good at time t to create a new firm with one product at time $t + 1$. Let $x_{x,t}^n$ denote the total amount of research goods purchased by $M_{x,t+1}^n$ entrants at time t , implying $x_{x,t}^n = M_{x,t+1}^n / M_{x,t}$.

Denote by $V_{x,t}(z_x)$ the value of a non-energy intermediate product with productivity z_x at time t . The free-entry condition is

$$\frac{1}{M_{x,t}} P_{x,t}^r = \mathbb{E}_t \frac{1}{1 + R_t} V_{x,t+1}(z'_x), \quad (\text{A.19})$$

where $P_{x,t}^r$ is the price of the research good. The left-hand side represents the marginal

cost of innovation for an entrant, while the right-hand side is the expected discounted value of the resulting product at $t + 1$.

Innovative investment by incumbent firms. Incumbent non-energy firms can invest in research goods either to create new products or to improve existing ones.

First, consider product creation. An incumbent firm can invest $x_{x,t}^m(z_x)$ units of the research good at time t to create a new product at $t + 1$ with probability $h(x_{x,t}^m(z_x)/s_{x,t}(z_x))$, where

$$s_{x,t}(z_x) \equiv \left(\frac{z_x}{Z_{x,t}} \right)^{\rho-1}. \quad (\text{A.20})$$

As with entrants, the productivity of the new product satisfies $\mathbb{E}z_x'^{\rho-1} = \eta_m z_x^{\rho-1}$. The aggregate research investment for product creation is $x_{x,t}^m = \sum_{z_x} M_{x,t}(z_x) x_{x,t}^m(z_x)$.

Second, consider productivity improvement. An incumbent firm can invest $x_{x,t}^c(z_x)$ units of the research good at time t to improve the productivity of an existing product. Conditional on survival, the new productivity z_x' is drawn such that $\mathbb{E}z_x'^{\rho-1} = \zeta (x_{x,t}^c(z_x)/s_{x,t}(z_x)) z_x^{\rho-1}$. The aggregate investment in this activity is $x_{x,t}^c = \sum_{z_x} M_{x,t}(z_x) x_{x,t}^c(z_x)$.

Under these assumptions, the intertemporal problem for a product with productivity index z_x can be written as

$$\begin{aligned} V_{x,t}(z_x) = & \max_{x_{x,t}^m(z_x), x_{x,t}^c(z_x)} \pi_{x,t}(z_x) (1 - \tau_t^{\text{Corp}}) - P_{x,t}^r(x_{x,t}^m(z_x) + x_{x,t}^c(z_x)) \\ & + \mathbb{E}_t \frac{V_{x,t+1}(z_x')}{1 + R_t} \left((1 - \delta_0) + h\left(\frac{x_{x,t}^m(z_x)}{s_{x,t}(z_x)}\right) \right), \end{aligned} \quad (\text{A.21})$$

where variable profits $\pi_{x,t}(z_x)$ are defined in (10). The continuation value, $V_{x,t+1}(z_x')$, is weighted by two components: $(1 - \delta_0)$, the probability of retaining an existing product, and $h(x_{x,t}^m(z_x)/s_{x,t}(z_x))$, the probability of creating a new product.

We solve Eq. (A.21) following Atkeson and Burstein (2019). First, variable profits scale with $s_{x,t}(z_x)$, so that $\pi_{x,t}(z_x) = s_{x,t}(z_x)(1 - \tau_t^{\text{Corp}})^{\frac{\mu-1}{\mu}} Y_{x,t}$. Second, innovative investment also scales with $s_{x,t}(z_x)$, implying $x_{x,t}^j(z_x) = s_{x,t}(z_x)x_{x,t}^j$ for $j \in \{c, m, n\}$. This yields $V_{x,t}(z_x) = V_{x,t}s_{x,t}(z_x)$, where $V_{x,t}$ satisfies:

$$V_{x,t} = \max_{x_{x,t}^m, x_{x,t}^c} (1 - \tau_t^{\text{Corp}})^{\frac{\mu-1}{\mu}} P_{x,t} Y_{x,t} - P_{x,t}^r (x_{x,t}^m + x_{x,t}^c) + \mathbb{E}_t \frac{V_{x,t+1}}{1 + R_t} ((1 - \delta_0) \zeta(x_{x,t}^c) + \eta_{x,m} h(x_{x,t}^m)) \frac{Z_{x,t}^{\rho-1}}{Z_{x,t+1}^{\rho-1}}. \quad (\text{A.22})$$

The first-order conditions for incumbent innovation and the free-entry condition for entrants are

$$x_{x,t}^m : \quad P_{x,t}^r = \frac{1}{1 + R_t} V_{x,t+1} \eta_{m,x} h'(x_{x,t}^m) \left(\frac{Z_{x,t}}{Z_{x,t+1}} \right)^{\rho-1}, \quad (\text{A.23})$$

$$x_{x,t}^c : \quad P_{x,t}^r = \frac{1 - \delta_0}{1 + R_t} V_{x,t+1} \zeta'(x_{x,t}^c) \left(\frac{Z_{x,t}}{Z_{x,t+1}} \right)^{\rho-1}, \quad (\text{A.24})$$

$$\text{Free-entry :} \quad P_{x,t}^r = \frac{1}{1 + R_t} V_{x,t+1} \eta_{n,x} \left(\frac{Z_{x,t}}{Z_{x,t+1}} \right)^{\rho-1}. \quad (\text{A.25})$$

Static problem: Energy intermediate good producers. The static problem of an energy intermediate firm with productivity index z_e consists of choosing dirty-energy inputs $\mathcal{D}_t(z_e)$, clean-energy inputs $\mathcal{C}_t(z_e)$, and price $p_e(z_e)$ to maximize variable profits (14), subject to the demand system (A.7) and the production technology, (11). The solution is characterized by the following conditions:

$$\frac{P_{\mathcal{D},t}}{P_{\mathcal{C},t}} = \left(\frac{\theta_e}{1 - \theta_e} \right)^{\frac{1}{\sigma_e}} \left(\frac{\mathcal{C}_t(z_e)}{\mathcal{D}_t(z_e)} \right)^{\frac{1}{\sigma_e}} \Gamma_{\mathcal{D},t}^* \quad (\text{A.26})$$

$$\frac{P_{\mathcal{D},t}}{p_{e,t}(z_e)} = \lambda_{e,t}(z_e) \theta_e^{\frac{1}{\sigma_e}} \left(\frac{\left[\theta_e^{\frac{1}{\sigma_e}} ((1 - \Gamma_{\mathcal{D},t}) \mathcal{D}_t(z_e))^{\frac{\sigma_e-1}{\sigma_e}} + (1 - \theta_e)^{\frac{1}{\sigma_e}} C_t(z_e)^{\frac{\sigma_e-1}{\sigma_e}} \right]^{\frac{\sigma_e}{\sigma_e-1}}}{\mathcal{D}_t(z_e)(1 - \Gamma_{\mathcal{D},t})} \right)^{\frac{1}{\sigma_e}} \Gamma_{\mathcal{D},t}^* \quad (\text{A.27})$$

$$\lambda_{e,t}(z_e) = \max \left\{ 0, \frac{1}{z_e} \text{MC}_{e,t} \right\}, \quad (\text{A.28})$$

$$p_{e,t}(z_e) = \mu \frac{1}{z_e} \text{MC}_{e,t}, \quad (\text{A.29})$$

$$\mu = \min \left\{ \frac{\rho}{\rho - 1}, \bar{\mu} \right\}, \quad (\text{A.30})$$

$$\text{MC}_{e,t} \equiv \left[\theta_e \left(\frac{P_{\mathcal{D},t}}{\Gamma_{\mathcal{D},t}^*} \right)^{1-\sigma_e} + (1 - \theta_e) P_{C,t}^{1-\sigma_e} \right]^{\frac{1}{1-\sigma_e}}, \quad (\text{A.31})$$

where we have defined $\Gamma_{\mathcal{D},t}^* = 1 - \Gamma_{\mathcal{D},t} - \mathcal{D}(z_e)_t \Gamma'_{\mathcal{D}}$.

Dynamic problem: Energy intermediate good producers. Let $M_{e,t+1}^n$ denote the measure of energy intermediate entrants that invest in innovation at time t . These firms draw productivity z'_e from a distribution such that $\mathbb{E}_t z_e'^{\rho-1} = \eta_n Z_{e,t}^{\rho-1} / M_{e,t}$. Aggregate productivity $Z_{e,t}$ and the total measure of energy products $M_{e,t}$ are defined as

$$Z_{e,t} = \left(\sum_{z_e} z_e^{\rho-1} M_{e,t}(z_e) \right)^{\frac{1}{\rho-1}} \quad (\text{A.32})$$

$$M_{e,t} = \sum_{z_e} M_{e,t}(z_e). \quad (\text{A.33})$$

An entering energy producer must purchase $1/M_{e,t}$ units of the energy research good at time t to create a new firm with one product at $t + 1$. Let $x_{e,t}^n$ denote total research investment by entrants at time t , implying $x_{e,t}^n = M_{e,t+1}^n/M_{e,t}$.

The free-entry condition is

$$\frac{1}{M_{e,t}} P_{e,t}^r = \mathbb{E}_t \frac{1}{1 + R_t} V_{e,t+1}(z_e'), \quad (\text{A.34})$$

where $P_{e,t}^r$ is the price of the energy research good, and $V_{e,t+1}(z_e')$ is the value of an energy intermediate good with productivity z_e' at $t + 1$, given by

$$V_{e,t}(z_e) = (1 - \tau_t^{\text{Corp}}) \pi_{e,t}(z_e) + \mathbb{E}_t \frac{1}{1 + R_t} V_{e,t+1}(z_e') (1 - \delta_0), \quad (\text{A.35})$$

where variable profits $\pi_{e,t}(z_e)$ are defined in (14). The continuation value reflects the probability $(1 - \delta_0)$ of retaining the product.

Following the same steps as for non-energy firms, $V_{e,t}(z_e)$ scales with $s_{e,t}(z_e)$, so that $V_{e,t}(z_e) = V_{e,t} s_{e,t}(z_e)$, where $V_{e,t}$ is

$$V_{e,t} = (1 - \tau_t^{\text{Corp}}) \frac{\mu - 1}{\mu} P_{e,t} Y_{e,t} + \mathbb{E}_t \frac{V_{e,t+1}}{1 + R_t} (1 - \delta_0) \frac{Z_{e,t}^{\rho-1}}{Z_{e,t+1}^{\rho-1}}. \quad (\text{A.36})$$

For energy intermediate firms, innovation decisions are governed solely by the free-entry condition.

$$\text{Free-entry : } P_{e,t}^r = \frac{1}{1 + R_t} V_{e,t+1} \eta_{n,e} \left(\frac{Z_{e,t}}{Z_{e,t+1}} \right)^{\rho-1}. \quad (\text{A.37})$$

A.2.3 Clean Energy Production

The maximization problem of the clean-energy producer (16) yields the following first-order condition:

$$r_t^C = P_{C,t} \Omega_C \alpha_e (K_t^C)^{\alpha_e - 1}. \quad (\text{A.38})$$

A.2.4 Research Good Producer

The maximization problem of the non-energy research good producer, (18), yields the following first-order condition for research labor demand:

$$P_{x,t}^R A_{x,t}^R Z_{x,t}^{\phi - 1} = W_t. \quad (\text{A.39})$$

Similarly, the first-order condition for the energy research good producer is

$$P_{e,t}^R A_{e,t}^R Z_{e,t}^{\phi - 1} = W_t. \quad (\text{A.40})$$

A.3 Aggregation and Productivity Dynamics

Under the assumptions on firm innovation and constant markups, aggregate non-energy output is

$$Y_{x,t} = Z_{x,t} (K_t)^\alpha (L_t^p)^{1-\alpha}, \quad (\text{A.41})$$

where aggregate productivity $Z_{x,t}$ is defined in (A.17).

Similarly, aggregate energy output is

$$Y_{e,t} = Z_{e,t} \left[\theta_e^{\frac{1}{\sigma_e}} ((1 - \Gamma_{\mathcal{D},t}) \mathcal{D}_t)^{\frac{\sigma_e - 1}{\sigma_e}} + (1 - \theta_e)^{\frac{1}{\sigma_e}} \mathcal{C}_t^{\frac{\sigma_e - 1}{\sigma_e}} \right]^{\frac{\sigma_e}{\sigma_e - 1}}, \quad (\text{A.42})$$

where $Z_{e,t}$ is defined in (A.32), and $\mathcal{C}_t$ and $\mathcal{D}_t$ denote aggregate clean- and dirty-energy use.

These expressions show that firm-level innovation—and policies affecting it—drive endogenous changes in aggregate productivity $Z_{x,t}$ and $Z_{e,t}$, which can be interpreted as measures of efficiency in non-energy and energy production.

This mechanism becomes clearer by deriving the dynamics of aggregate productivity, following Atkeson and Burstein (2019). We begin with the evolution of the total measure of products in the non-energy and energy sectors, $M_{x,t}$ and $M_{e,t}$, given by

$$M_{x,t+1} = (1 - \delta_0)M_{x,t} + x_{x,t}^n M_{x,t} + h(x_{x,t}^m) M_{x,t}, \quad (\text{A.43})$$

$$M_{e,t+1} = (1 - \delta_0)M_{e,t} + x_{e,t}^n M_{e,t}. \quad (\text{A.44})$$

These expressions imply that the total measure of products at $t + 1$ is driven by three forces. First, exogenous exit reduces the mass of existing products, so only a fraction, $1 - \delta_0$, survives from period t . Second, entry by new firms— $M_{x,t+1}^n = x_{x,t}^n M_{x,t}$ and $M_{e,t+1}^n = x_{e,t}^n M_{e,t}$ —adds new products created through innovation at time t . Third, for non-energy goods, incumbent firms contribute additional product creation through innovative investment.

Following a similar approach, the dynamics of aggregate productivity for non-energy and energy goods are

$$Z_{x,t+1}^{\rho-1} = (1 - \delta_0)\zeta(x_{x,t}^c)M_{x,t} \frac{Z_{x,t}^{\rho-1}}{M_{x,t}} + \eta_n x_{x,t}^n M_{x,t} \frac{Z_{x,t}^{\rho-1}}{M_{x,t}} + \eta_m h(x_{x,t}^m)M_{x,t} \frac{Z_{x,t}^{\rho-1}}{M_{x,t}} \quad (\text{A.45})$$

$$Z_{e,t+1}^{\rho-1} = (1 - \delta_0)M_{e,t} \frac{Z_{e,t}^{\rho-1}}{M_{e,t}} + \eta_n x_{e,t}^n M_{e,t} \frac{Z_{e,t}^{\rho-1}}{M_{e,t}}. \quad (\text{A.46})$$

Next-period productivity reflects three components. The first term captures the average productivity of surviving incumbent products. The second term reflects the contri-

bution of new products created by entrants. The third term—present only for non-energy goods—captures productivity gains from incumbent innovation.

Taking logs of (A.45) and rearranging yields productivity growth rates $g_{Z,x,t} \equiv \log Z_{x,t+1} - \log Z_{x,t}$ and $g_{Z,e,t} \equiv \log Z_{e,t+1} - \log Z_{e,t}$:

$$g_{Z,x,t} = \frac{1}{\rho - 1} \log \left((1 - \delta_0) \zeta(x_{x,t}^c) + \eta_m h(x_{x,t}^m) + \eta_n x_{x,t}^n \right), \quad (\text{A.47})$$

$$g_{Z,e,t} = \frac{1}{\rho - 1} \log \left((1 - \delta_0) + \eta_n x_{e,t}^n \right), \quad (\text{A.48})$$

which makes explicit how productivity growth depends on innovation by incumbents and entrants.

A.4 BGP

The model features endogenous growth; hence, we detrend variables by their BGP rates to obtain a stationary equilibrium.

Let lowercase variables denote detrended quantities. For example, for output Y_t , define $y_t \equiv Y_t / \exp(tg_y)$, where g_y is the constant BGP growth rate. This variable is constant along the BGP. Using this notation, the equilibrium system can be expressed in stationary form as

$$\frac{1}{c_t} = \mathbb{E}_t \beta (1 + R_t) \exp(g_Y - g_H) \frac{1}{c_{t+1}}, \quad (\text{A.49})$$

$$\frac{1}{c_t} = \mathbb{E}_t \exp(g_Y - g_H) \frac{1}{c_{t+1}} \frac{1}{q_t} \left(r_{t+1}^k (1 - \tau_{t+1}^{\text{Corp}}) + q_{t+1} \left(1 - \delta_K - \frac{\partial \Xi^I(i_{t+1}, k_{t+1})}{\partial k_{t+1}} \right) \right), \quad (\text{A.50})$$

$$\frac{1}{q_t} = 1 - \frac{\partial \Xi^I(i_t, k_t)}{\partial i_t}, \quad (\text{A.51})$$

$$\frac{1}{c_t} = \mathbb{E}_t \exp(g_Y - g_H) \frac{1}{c_{t+1}} \frac{1}{q_t^C} \left(r_{t+1}^C (1 - \tau_{t+1}^{\text{Corp}}) + q_{t+1}^C \left(1 - \delta_K - \frac{\partial \Xi^I(i_{t+1}^C, k_{t+1}^C)}{\partial k_{t+1}^C} \right) \right), \quad (\text{A.52})$$

$$\frac{1}{q_t^C} = 1 - \frac{\partial \Xi^I(i_t^C, k_t^C)}{\partial i_t^C}, \quad (\text{A.53})$$

$$\left(\frac{l_t}{h_t} \right)^\varphi \frac{w_t}{h_t} \exp(g_Y - g_H) = (1 - \tau_t^L) w_t \exp(g_Y - g_H),$$

$$y_t = \left[\theta^{\frac{1}{\sigma}} y_{x,t}^{\frac{\sigma-1}{\sigma}} + (1 - \theta)^{\frac{1}{\sigma}} y_{x,t}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (\text{A.54})$$

$$p_{x,t} = \theta^{\frac{1}{\sigma}} \left(\frac{y_t}{y_{x,t}} \right)^{\frac{1}{\sigma}}, \quad (\text{A.55})$$

$$p_{e,t} = (1 - \theta)^{\frac{1}{\sigma}} \left(\frac{y_t}{y_{e,t}} \right)^{\frac{1}{\sigma}}, \quad (\text{A.56})$$

$$y_{e,t} = z_{e,t} \left[\theta_e^{\frac{1}{\sigma_e}} ((1 - \Gamma_{\mathcal{D},t}) \vartheta_t)^{\frac{\sigma_e-1}{\sigma_e}} + (1 - \theta_e)^{\frac{1}{\sigma_e}} c_t^{\frac{\sigma_e-1}{\sigma_e}} \right]^{\frac{\sigma_e}{\sigma_e-1}}, \quad (\text{A.57})$$

$$\frac{p_{\mathcal{D},t}}{p_{e,t}} = \frac{1}{\mu} z_{e,t} \theta_e^{\frac{1}{\sigma_e}} \left(\frac{\left[\theta_e^{\frac{1}{\sigma_e}} ((1 - \Gamma_{\mathcal{D},t}) \mathcal{D})_t^{\frac{\sigma_e-1}{\sigma_e}} + (1 - \theta_e)^{\frac{1}{\sigma_e}} c_t^{\frac{\sigma_e-1}{\sigma_e}} \right]^{\frac{\sigma_e}{\sigma_e-1}}}{(1 - \Gamma_{\mathcal{D},t}) \mathcal{D}_t} \right)^{\frac{1}{\sigma_e}} \Gamma_{\mathcal{D},t}^* \quad (\text{A.58})$$

$$\frac{p_{C,t}}{p_{e,t}} = \frac{1}{\mu} z_{e,t} (1 - \theta_e)^{\frac{1}{\sigma_e}} \left(\frac{\left[\theta_e^{\frac{1}{\sigma_e}} ((1 - \Gamma_{\mathcal{D},t}) \mathcal{D})_t^{\frac{\sigma_e-1}{\sigma_e}} + (1 - \theta_e)^{\frac{1}{\sigma_e}} c_t^{\frac{\sigma_e-1}{\sigma_e}} \right]^{\frac{\sigma_e}{\sigma_e-1}}}{c_t} \right)^{\frac{1}{\sigma_e}}, \quad (\text{A.59})$$

$$c_t = \Omega_C (k_t^c)^{\alpha_e}, \quad (\text{A.60})$$

$$r_t^c = \alpha_e p_{C,t} \Omega_C (k_t^c)^{\alpha_e-1}, \quad (\text{A.61})$$

$$y_{x,t} = z_{x,t} (k_t)^\alpha (l_t^p)^{1-\alpha}, \quad (\text{A.62})$$

$$\frac{w_t}{p_{x,t}} = \frac{1 - \alpha}{\mu} \frac{y_{x,t}}{l_t^p}, \quad (\text{A.63})$$

$$\frac{r_t^K}{p_{x,t}} = \frac{\alpha}{\mu} \frac{y_{x,t}}{k_t}, \quad (\text{A.64})$$

$$p_{x,t}^R \gamma_{x,t}^R = w_t l_{x,t}^R, \quad (\text{A.65})$$

$$p_{e,t}^R \gamma_{e,t}^R = w_t l_{e,t}^R, \quad (\text{A.66})$$

$$\begin{aligned}
v_{x,t} = & \max_{x_{x,t}^m, x_{x,t}^c} (1 - \tau_t^{\text{Corp}}) \frac{\mu - 1}{\mu} - p_{x,t}^r (x_{x,t}^m + x_{x,t}^c) \\
& + \mathbb{E}_t \exp(g_Y - (\rho - 1)g_{z,x}) \frac{v_{x,t+1}}{1 + R_t} ((1 - \delta_0)\zeta(x_{x,t}^c) + \eta_{m,x}h(x_{x,t}^m)) \frac{z_{x,t}^{\rho-1}}{z_{x,t+1}^{\rho-1}} \frac{p_{x,t+1}y_{x,t+1}}{p_{e,t}y_{e,t}},
\end{aligned} \tag{A.67}$$

$$p_{x,t}^r = \frac{\exp(g_Y - (\rho - 1)g_{z,x})}{1 + R_t} v_{x,t+1} \eta_{n,x} \left(\frac{z_{x,t}}{z_{x,t+1}} \right)^{\rho-1} \frac{p_{x,t+1}y_{x,t+1}}{p_{x,t}y_{x,t}}, \tag{A.68}$$

$$p_{x,t}^r = \frac{\exp(g_Y - (\rho - 1)g_{z,x})}{1 + R_t} v_{x,t+1} (1 - \delta_0) \zeta'(x_{x,t}^c) \left(\frac{z_{x,t}}{z_{x,t+1}} \right)^{\rho-1} \frac{p_{x,t+1}y_{x,t+1}}{p_{x,t}y_{x,t}}, \tag{A.69}$$

$$p_{x,t}^r = \frac{\exp(g_Y - (\rho - 1)g_{z,x})}{1 + R_t} v_{x,t+1} \eta_m h'(x_{x,t}^m) \left(\frac{z_{x,t}}{z_{x,t+1}} \right)^{\rho-1}, \tag{A.70}$$

$$\begin{aligned}
v_{e,t} = & (1 - \tau_t^{\text{Corp}}) \frac{\mu - 1}{\mu} \\
& + \mathbb{E}_t \exp(g_Y - (\rho - 1)g_{z,e}) \frac{v_{e,t+1}}{1 + R_t} (1 - \delta_0) \frac{z_{e,t}^{\rho-1}}{z_{e,t+1}^{\rho-1}} \frac{p_{e,t+1}y_{e,t+1}}{p_{e,t}y_{e,t}},
\end{aligned} \tag{A.71}$$

$$p_{e,t}^r = \frac{\exp(g_Y - (\rho - 1)g_{z,e})}{1 + R_t} v_{e,t+1} \eta_{n,e} \left(\frac{z_{e,t}}{z_{e,t+1}} \right)^{\rho-1} \frac{p_{e,t+1}y_{e,t+1}}{p_{e,t}y_{e,t}}, \tag{A.72}$$

$$Y_{x,t}^r = a_{x,t} z_{x,t}^{\phi-1} l_{x,t}^r, \tag{A.73}$$

$$Y_{e,t}^r = a_{e,t} z_{e,t}^{\phi-1} l_{e,t}^r, \quad (\text{A.74})$$

$$\left(\frac{\exp(g_{z,x}) z_{x,t+1}}{z_{x,t}} \right)^{\rho-1} = (1 - \delta_0) \zeta(x_{x,t}^c) + \eta_{m,x} h(x_{x,t}^m) + \eta_{n,x} x_{x,t}^n, \quad (\text{A.75})$$

$$\left(\frac{\exp(g_{z,x}) z_{x,t+1}}{z_{x,t}} \right)^{\rho-1} = (1 - \delta_0) + \eta_{n,e} x_{e,t}^n, \quad (\text{A.76})$$

$$Y_{x,t}^r = x_{x,t}^n + x_{x,t}^m + x_{x,t}^c, \quad (\text{A.77})$$

$$Y_{e,t}^r = x_{e,t}^n, \quad (\text{A.78})$$

$$c_t + i_t + i_t^c = y_t - p_{\mathcal{D},t} d_t - \Xi[i_t, k_t] - \Xi[i_t^c, k_t^c]. \quad (\text{A.79})$$

A.5 Calibration of Firm Innovation Process

As indicated in the main text, our calibration strategy for firm innovation follows [Domínguez-Díaz et al. \(2025\)](#), which closely follows [Atkeson and Burstein \(2019\)](#). Because the data do not clearly distinguish between non-energy and energy-oriented innovation by final use, we assume that they primarily reflect non-energy innovation and match the corresponding model moments to the data. As a sanity check, we then assess the model's energy-related innovation outcomes. The calibration delivers reasonable BGP aggregates: energy-related innovation accounts for 4.94% of total innovation, close to the model's aggregate energy-to-GDP ratio of 4.8%.

We next outline the calibration procedure. First, we posit the following functional

forms for incumbent innovation in new products $h(x_{x,t}^m)$ and existing products $\zeta(x_{x,t}^c)$ among non-energy producers:

$$h(x_{x,t}^m) = h_0(x_{x,t}^m)^{h_1} \quad (\text{A.80})$$

$$\zeta(x_{x,t}^c) = \zeta_0 + \zeta_1(x_{x,t}^c)^{\zeta_2}. \quad (\text{A.81})$$

Eight parameters remain to be calibrated: productivity steps for entrant and incumbent new products, η_n and η_m ; the exogenous product exit rate, δ_0 ; and parameters governing incumbent innovation in new products (h_0 and h_1) and existing products (ζ_0 , ζ_1 , and ζ_2).

Our strategy follows [Atkeson and Burstein \(2019\)](#). Let f_n denote the fraction of products produced by entrants (M_x^n/M_x), and let f_m and $f_c = 1 - f_n - f_m$ denote the corresponding fractions of new and continuing products produced by incumbents. Let s_n denote the aggregate size of products produced by entrants, defined by aggregating Eq. (A.20), and define s_m and $s_c = 1 - s_n - s_c$ analogously for new and continuing incumbent products. The average size of each product type is $\text{avsize}_j = s_j/f_j$ for $j \in \{n, m, c\}$. Finally, let g_{M_x} denote the growth rate of M_x .

The parameter values for $\{\delta_0, \eta_m, \eta_n\}$ and the steady-state values of $\{h(x_x^m), \zeta(x_x^c), x_x^n\}$

are pinned down by:

$$h(x_x^m) = f_m \exp(g_{M_x}), \quad (\text{A.82})$$

$$x_x^n = f_n \exp(g_{M_x}), \quad (\text{A.83})$$

$$\zeta(x_x^c) = \text{avsize}_c \frac{\exp((\rho - 1)g_{Z_x})}{\exp g_{M_x}}, \quad (\text{A.84})$$

$$\eta_m = \text{avsize}_m \frac{\exp((\rho - 1)g_{Z_x})}{\exp g_{M_x}}, \quad (\text{A.85})$$

$$\eta_n = \text{avsize}_e \frac{\exp((\rho - 1)g_{Z_x})}{\exp g_{M_x}}, \quad (\text{A.86})$$

$$\delta_0 = 1 - f_c \exp(g_{M_x}). \quad (\text{A.87})$$

These equations are derived from the law of motion for M_x (A.43), the law of motion for Z_x (A.45), the expressions for the total measure of new products created by entrants and incumbents ($x_{x,t}^n = M_{x,t+1}^n / M_{x,t}$ and $h(x_x^m) = M_{x,t+1}^m / M_{x,t}$), and the assumed productivity distributions for new products created by entrants ($\mathbb{E} z_x^{\rho-1} = \eta_n Z_{x,t}^{\rho-1} / M_{x,t}$) and incumbents ($\mathbb{E} z_x^{\rho-1} = \eta_m z_x^{\rho-1}$).

Given these objects, we use firms' first-order conditions to obtain $h'(x_x^m)$ and $\zeta'(x_x^c)$, as derived in Appendix A:

$$\eta_n = \eta_m h'(x_x^m) \quad (\text{A.88})$$

$$\eta_n = (1 - \delta_0) \zeta'(x_x^c). \quad (\text{A.89})$$

Note that x_x^n is already determined by equation (A.83). Given this, we resolve $x_x^c + x_x^m$ by targeting an aggregate research and development (R&D)-to-GDP ratio of 1.41%, based on INE data. This condition jointly determines x_x^c and x_x^m , but not their individual values. To separate them, we follow Atkeson and Burstein (2019). Specifically, the ratio, $P_x^r x_x^c / P_x Y_x$, must lie between an upper bound, $P_x^r (x_x^c + x_x^m) / P_x Y_x$, and a lower bound obtained by

evaluating firm value when innovation in continuing products is zero.¹⁰ We set $P_x^r x_x^c / P_x Y_x$ at the midpoint of these bounds, which identifies x_x^c . Given x_x^c and x_x^n , this also determines x_x^m .

Given x_x^m , together with $h(x_x^m)$ and $h'(x_x^m)$, we recover h_0 and h_1 using (A.80). We set $\zeta_2 = 0.5$, the midpoint of admissible values in Atkeson and Burstein (2019). Given ζ_2 and x_x^c , we then recover ζ_0 and ζ_1 from (A.81), using the previously obtained values of $\zeta(x_x^c)$ and $\zeta'(x_x^c)$.

To implement this calibration, we use two firm-level data sources in addition to aggregate R&D expenditures. First, we use the Central Business Directory of INE (*DIRCE*) to obtain a firm growth rate $g_{M,x}$ of approximately 1%. Second, we use firm-level data from the Central Balance Sheet Data Office. While Atkeson and Burstein (2019) relied on employment and establishment counts, our data contain only firm-level information, so we use production-based measures instead. The resulting moments are nonetheless close to those reported in Table 3 of their online appendix.

From these data, we measure $F_n = 0.02$ as the share of production accounted for by new firms, $F_m = 0.05$ as the share due to expansion by incumbent firms, and $F_c = 0.93$ from $F_c = 1 - F_n - F_m$. We also measure $S_n = 0.03$ as the employment share of new firms, $S_m = 0.03$ as the share from incumbent expansion, and $S_c = 0.94$ from $S_c = 1 - S_n - S_m$.

B Carbon Taxes and Green Transition: Counterfactual Economies

In the main text (Section 5.1), we showed that carbon taxes promote a greener economy in the long run by stimulating clean-energy investment and improving energy efficiency. In this appendix, we quantify the underlying mechanisms by analyzing a set of counterfactual economies.

¹⁰A derivation of these bounds can be found in Appendix C.3 of Atkeson and Burstein (2019).

	Clean energy Share	Dirty Energy over GDP	Total energy over GDP	GDP	Clean energy Price	Dirty energy Price (after tax)	Energy efficiency	Non-energy Efficiency
	(%, level)	(%, level)	(%, level)	(%, change)	(%, change)	(%, change)	(%, change)	(%, change)
Medium clean energy share ($\tau^D = 0.4$)								
Medium clean energy Share	45.09	2.32	4.65	-1.38	5.37	40.00	6.28	-0.36
Revenue recycling	45.00	2.32	4.65	0.19	5.49	40.00	6.58	0.00
No energy efficiency Investment	44.76	2.46	4.62	-1.98	5.83	40.00	0.00	-0.40
No clean capital Investment	29.83	2.83	4.64	-2.32	31.21	40.00	8.27	-0.40
High clean energy share ($\tau^D = 2$)								
High clean energy Share	86.31	0.65	4.39	-3.50	14.46	200.00	17.22	-0.86
Revenue recycling	86.24	0.65	4.39	-1.28	14.68	200.00	17.72	-0.34
No energy efficiency Investment	85.88	0.76	4.30	-5.49	15.81	200.00	0.00	-1.04
No clean capital Investment	41.08	1.86	4.27	-9.31	138.43	200.00	29.40	-1.51

Notes: Top panel: aggregate effects of a carbon tax $\tau^D = 0.4$ across counterfactual economies. Bottom panel: aggregate effects of $\tau^D = 2$. The first row in each panel reports the baseline results from Table 2 for reference. See the main text for a detailed description of each counterfactual.

Table B.1: Long-run effects of higher carbon tax in counterfactual economies

The first counterfactual considers revenue recycling, whereby additional carbon-tax revenues are used to reduce the labor income tax rate, τ^L . The second isolates the role of endogenous innovation in energy efficiency by fixing Z_e , preventing it from responding to higher carbon taxes. The third counterfactual fixes clean capital K^C , highlighting the role of endogenous clean-energy production.

Table B.1 reports the macroeconomic effects of the green transition across counterfactual economies. The top panel corresponds to the carbon tax required to achieve the medium clean-energy share, while the bottom panel reports results for the high clean-energy share. The first row in each panel reproduces the baseline results from Table 2 for comparison.

Recycling carbon-tax revenues through lower labor income taxes (“Revenue Recycling”) substantially mitigates GDP losses. As shown in Column 4, GDP rises by 0.19% instead of falling by 1.38% when $\tau^D = 0.4$, and the GDP decline is reduced by nearly

two-thirds when $\tau^D = 2$. The mechanism operates through increased labor supply, which raises employment, production, and output. However, revenue recycling has little effect on the energy mix or energy ratios (Columns 1–3), indicating that these are primarily driven by relative energy price changes induced by the carbon tax.

The “No energy efficiency investment” counterfactual shows that endogenous innovation in energy efficiency is a key mitigation channel. Without it, GDP losses increase by nearly 60%, reaching -1.98% in the medium clean-energy case and -5.49% in the high clean-energy case (Column 4). Absent efficiency gains, firms must increase energy expenditures to sustain production, raising costs and reducing investment and labor demand. As in the previous case, relative energy prices—and thus energy ratios—are only marginally affected.

The “No clean capital investment” counterfactual shows that removing clean-energy investment sharply amplifies GDP losses, up to -2.32% (medium clean share) and -9.31% (high clean share). This reflects two mechanisms: clean-energy production contributes directly to value added, and a fixed supply of clean energy leads to a stronger increase in its price (see Column 5). Higher energy costs further depress non-energy investment and labor demand. Moreover, without clean capital adjustment, the carbon tax becomes insufficient to achieve the targeted transition: for example, in the high clean share case, the dirty-energy-to-GDP ratio falls only to 1.86% (instead of 0.65%), and the clean-energy share reaches only 41.08% (instead of 86%).

C Green Transition and Vulnerability to External Shocks:

Additional Results

In this appendix, we present complementary results to the main text. First, we extend the baseline model to include trade in non-energy goods. Second, we replicate the analysis

under a clean-input subsidy used to achieve the desired clean-energy share instead of a carbon tax. Finally, we show that the main results are robust to alternative calibration choices.

C.1 Trade in Non-Energy Goods

The baseline economy is a small open economy that imports dirty energy for production. In this section, we extend the model to allow for trade in non-energy goods. Specifically, we introduce a foreign economy that demands X_t goods from the domestic economy:

$$X_t = (P_{\mathcal{D},t}^*)^{\sigma_X} Y_t^*, \quad (\text{A.90})$$

where $P_{\mathcal{D},t}^*$ denotes the real price of energy in terms of domestic goods. From the perspective of the foreign economy, this corresponds to the price of its exports (energy) relative to its imports (goods produced in the small open economy). The parameter, σ_X , is the elasticity of substitution between domestic and foreign non-energy goods, and Y_t^* denotes foreign GDP. Similar specifications of foreign demand are standard in the open economy literature; see, e.g., [Galí and Monacelli \(2016\)](#).

Under this assumption, the aggregate resource constraint in (22) becomes

$$C_t + I_t + I_t^{\mathcal{C}} = Y_t + (X_t - P_{\mathcal{D},t}^* \mathcal{D}_t) - \Xi^I(I_t^{\mathcal{C}}, K_t^{\mathcal{C}}) - \Xi^I(I_t, K_t). \quad (\text{A.91})$$

Following [Gornemann et al. \(2022\)](#), we assume that foreign output Y_t^* co-moves one-to-one with the energy required to meet domestic demand, so that $Y_t^* = \mathcal{D}_t$. We also set the trade elasticity to unity, $\sigma_X = 1$, implying balanced trade in equilibrium. This specification provides a useful benchmark relative to the baseline without non-energy trade, as intermediate cases—where trade partially offsets energy imports—would lie between

these extremes. Such balanced-trade benchmarks are standard in the literature (e.g., [Gali and Monacelli \(2005\)](#)).

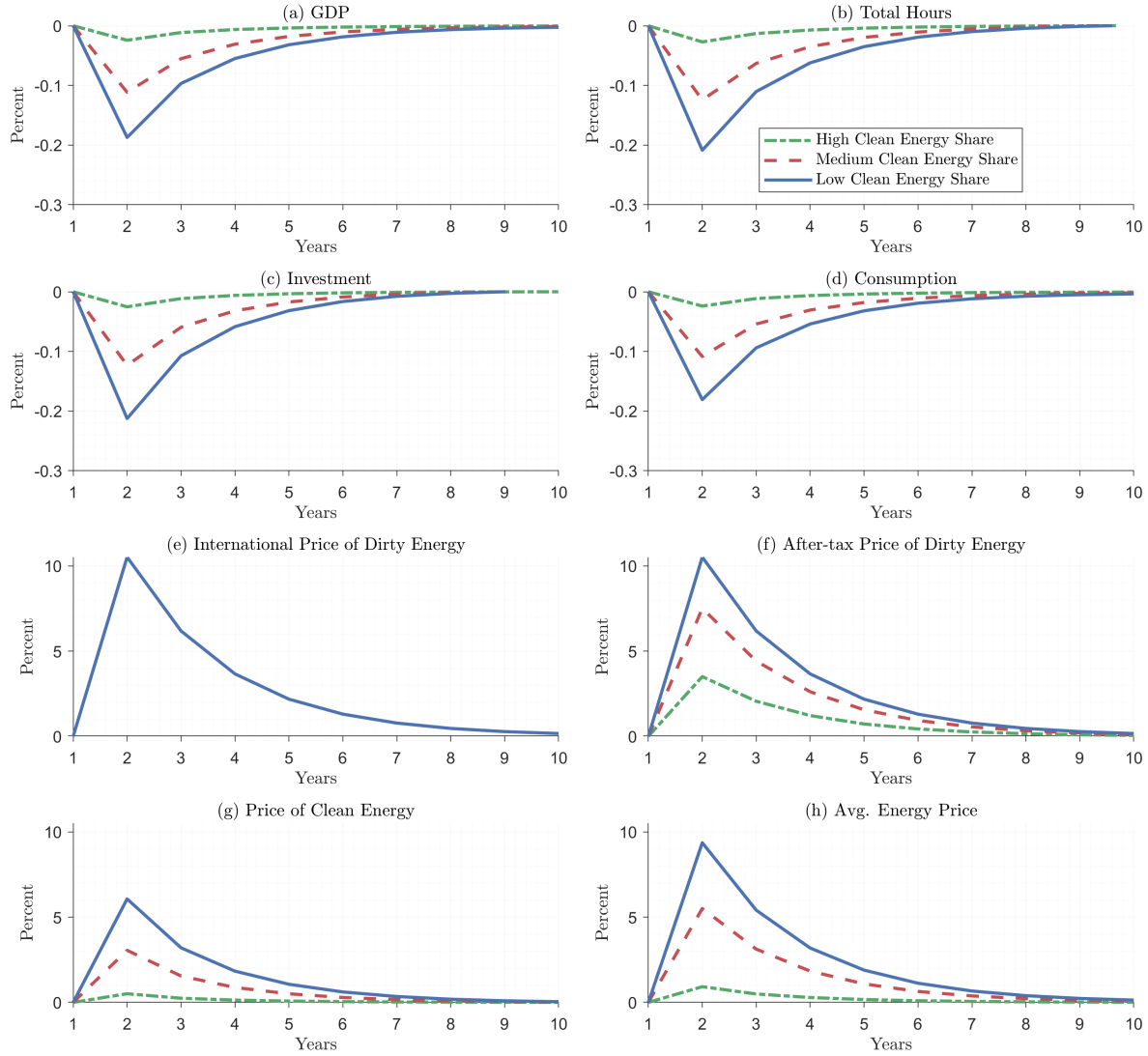
Figure [C.1](#) reports the responses to a 10% increase in the international price of dirty energy (Panel e) across low, medium, and high clean-energy share economies when trade in non-energy goods is allowed. The qualitative results mirror those in the main text (Fig. [1](#)): higher clean-energy shares reduce vulnerability to external energy price shocks, with smaller declines in GDP, consumption, investment, and employment. However, the magnitude of these effects is attenuated. Trade in non-energy goods introduces an additional adjustment channel, as higher energy prices increase foreign income, boost exports, and partially offset the decline in domestic output.

Importantly, the central result remains unchanged: increasing the clean-energy share from 26% to 86% sharply reduces vulnerability to international energy price shocks. In this setting, the impact on GDP in the high clean-energy economy (-0.025%) is about 13% of that in the low clean-energy economy (-0.19%), closely matching the baseline result (approximately 18%: -0.08% versus -0.43%).

Figure [C.2](#) reinforces this conclusion by showing the change in GDP volatility (standard deviation) in the medium and high clean-energy economies relative to the low clean-energy baseline. The decline in volatility remains substantial: GDP volatility in the high clean-energy economy falls by about 85%, closely aligning with the baseline results (see Fig. [2](#)).

C.2 Clean-input subsidy

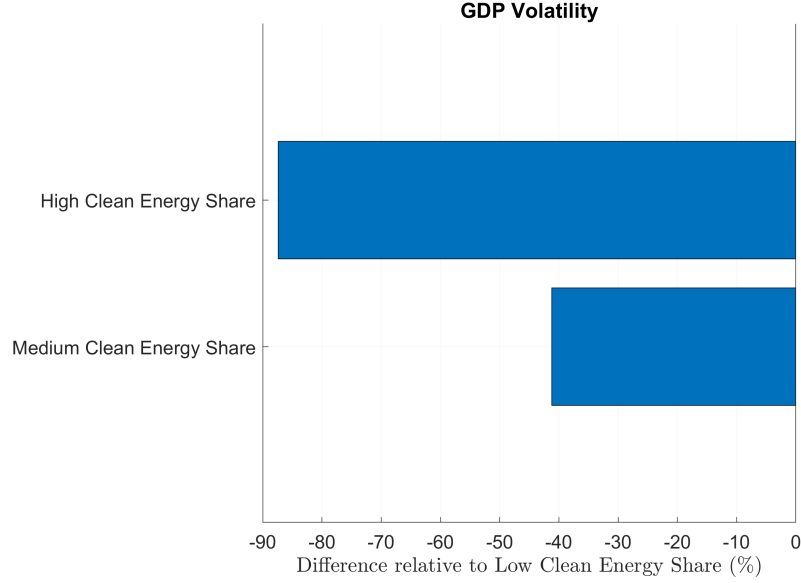
In the baseline analysis, the increase in the clean-energy share is driven by higher carbon taxes. An alternative policy instrument is a subsidy to clean-energy inputs. In this section, we examine whether the main results hold when a clean-input subsidy replaces the carbon tax in achieving the desired energy mix.



Notes: Impulse response functions to a 10% increase in the international price of dirty energy (Panel e). Blue solid line: low clean-energy share economy. Red dashed line: medium clean-energy share economy. Green dotted line: high clean-energy share economy. Increases in the clean-energy share are achieved through higher carbon taxes. The model includes trade in non-energy goods.

Figure C.1: Effects of an increase in the international price of dirty energy—Trade in non-energy goods

Specifically, we assume that the domestic clean-energy producer receives a per-unit subsidy, s_t . Under this assumption, the profit maximization problem in (16) becomes



Notes: Change in GDP volatility (standard deviation) in the medium and high clean-energy economies relative to the low clean-energy baseline. The standard deviation is computed from simulations of a long series of shocks to the international price of dirty energy. The model includes trade in non-energy goods.

Figure C.2: GDP volatility—Trade in non-energy good)

$$\max_{K_t^C} (1 + s_t)P_{C,t}C_t - r_t^C K_t^C \quad \text{s.t.} \quad C_t = \Omega_C \left(K_t^C \right)^{\alpha_C}. \quad (\text{A.92})$$

The subsidy directly promotes clean-energy production and lowers its price, increasing the clean-energy share in the economy. In the absence of a carbon tax, we calibrate the subsidy, s_t , in each scenario (low, medium, and high clean-energy share) to match the targets in the main text (26%, 56%, and 86%, respectively). Specifically, the subsidy is set to 0 in the low clean-energy economy, 0.423 in the medium scenario, and 2.2 in the high scenario.

Although a clean-input subsidy and a carbon tax can achieve the same energy mix, their fiscal implications differ substantially. The subsidy required to reach the medium clean-energy share entails a fiscal cost of about 0.7% of GDP, rising to about 5% for the high clean-energy share. By contrast, the carbon tax generates revenues: approximately

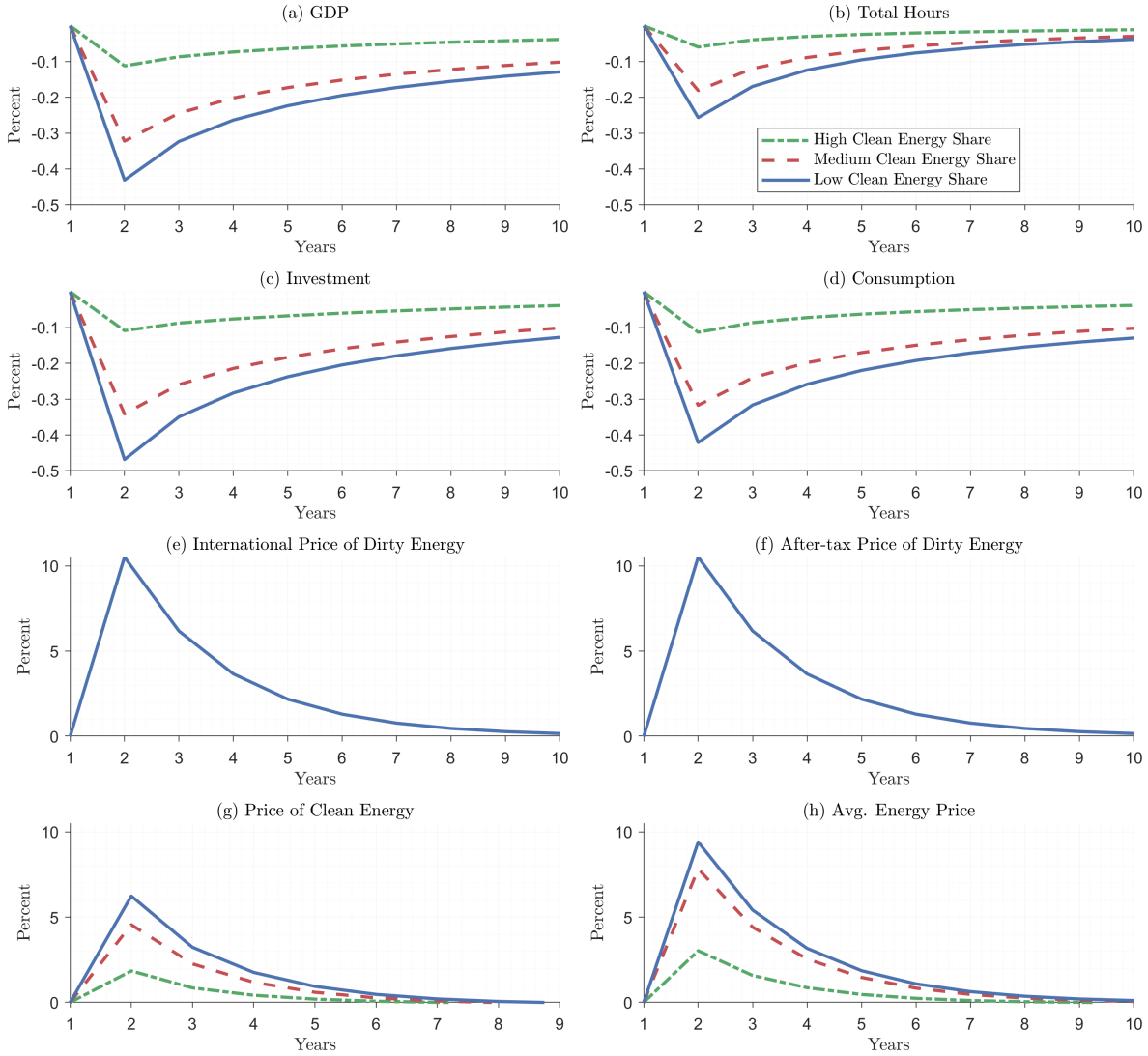
0.93% of GDP in the medium case and 0.26% in the high case. Despite the higher tax rate in the latter, the decline in dirty-energy use reduces the tax base and lowers total revenues.

We then analyze the transmission of a 10% increase in the international price of dirty energy (Panel e) across the three scenarios when the clean-energy share is achieved via a subsidy rather than a carbon tax. Figure C.3 reports the impulse responses, and Fig. C.4 shows the corresponding changes in GDP volatility in the medium and high clean-energy economies relative to the low clean-energy baseline.

Overall, the main results remain both quantitatively and qualitatively robust under this alternative policy instrument. As the clean-energy share rises, the economy becomes less vulnerable to external energy price shocks, with smaller declines in GDP, consumption, investment, and employment. In contrast to the baseline, the absence of a carbon tax implies that the international price of dirty energy (Panel e) coincides with the domestic price (Panel f). Nevertheless, the clean-input subsidy still facilitates partial decoupling of clean-energy prices (Panel g) from fluctuations in dirty-energy prices, thereby limiting the increase in the average energy price (Panel h). As a result, together with the reduced reliance on dirty energy, the impact on GDP (Panel a) and other macroeconomic variables diminishes as the clean-energy share increases, consistent with the main text.

C.3 Robustness

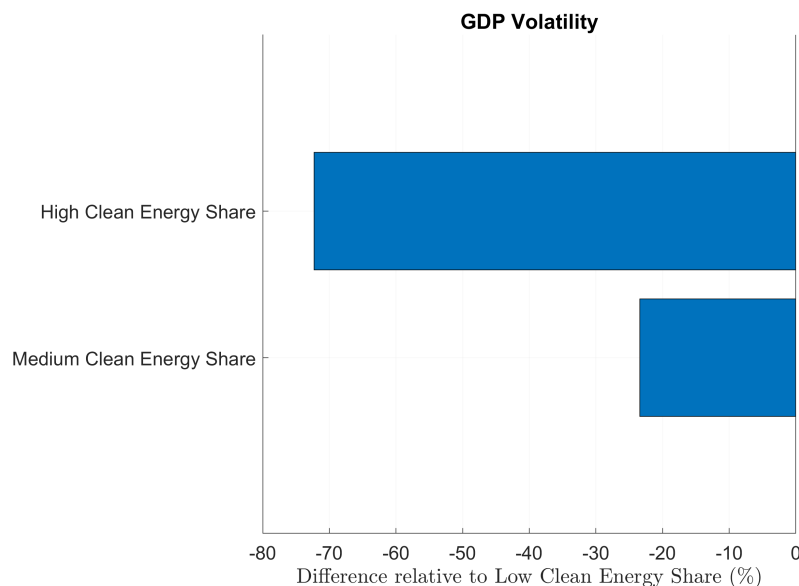
We consider the following alternative calibration choices to assess the robustness of our main results. First, we lower the elasticity of substitution between clean and dirty energy. Second, we eliminate short-run adjustment costs in substitution between these energy inputs. Third, we consider alternative parameters governing the innovation process.



Notes: Impulse response functions to a 10% increase in the international price of dirty energy (Panel e). Blue solid line: low clean-energy share economy. Red dashed line: medium clean-energy share economy. Green dotted line: high clean-energy share economy. Increases in the clean-energy share are achieved through a clean-input subsidy rather than carbon taxes.

Figure C.3: Effects of an increase in the international price of dirty energy—Clean-input subsidy

Elasticity of substitution between clean and dirty energy. In the baseline, the elasticity of substitution between clean and dirty energy is set to 3, following [Papageorgiou et al. \(2017\)](#) for non-energy industries. Given uncertainty in these estimates, we re-run the analysis with a lower elasticity of two, consistent with estimates for the electricity-generating

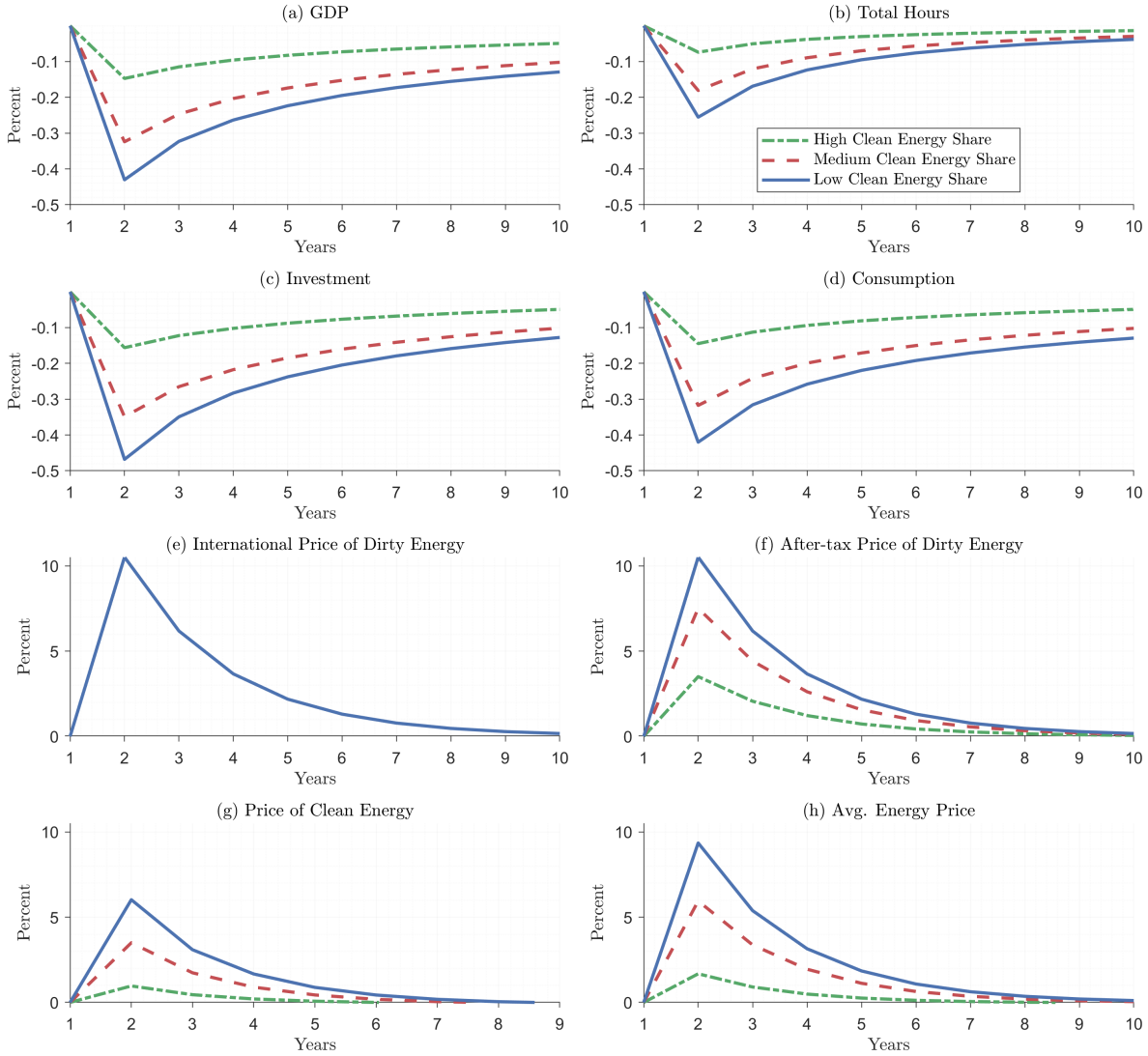


Notes: Change in GDP volatility (standard deviation) in the medium and high clean-energy economies relative to the low clean-energy baseline. The standard deviation is computed from simulations of a long series of shocks to the international price of dirty energy. Increases in the clean-energy share are achieved through a clean-input subsidy rather than carbon taxes.

Figure C.4: GDP volatility—Clean-input subsidy

sector in [Papageorgiou et al. \(2017\)](#).

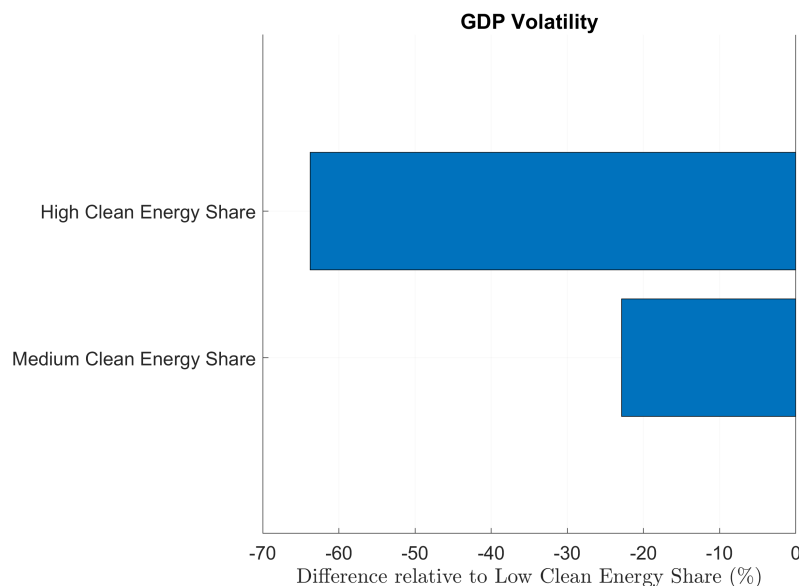
Figure C.5 reports the responses to a 10% increase in the international price of dirty energy (Panel e) across low, medium, and high clean-energy share economies under this alternative calibration. The overall impact of the shock is larger than in the baseline, as a lower elasticity reduces firms' ability to substitute away from dirty energy when its price rises. This increases production costs and leads to larger declines in GDP, consumption, investment, and employment. Despite this amplification, the main result remains robust: higher clean-energy shares reduce vulnerability to external energy price shocks. In particular, the impact on GDP declines markedly as the clean-energy share rises, as shown in Panel a of Fig. C.5 and in the corresponding reductions in GDP volatility reported in Fig. C.6.



Notes: Impulse response functions to a 10% increase in the international price of dirty energy (Panel e). Blue solid line: low clean-energy share economy. Red dashed line: medium clean-energy share economy. Green dotted line: high clean-energy share economy. Increases in the clean-energy share are achieved through higher carbon taxes. Robustness check with a lower elasticity of substitution between clean and dirty energy.

Figure C.5: Effects of an increase in the international price of dirty energy—Robustness: Lower elasticity of substitution between clean and dirty energy)

Short-run adjustment cost between clean and dirty energy. The baseline model includes short-run adjustment costs in substituting between clean and dirty energy, consistent with [Coenen et al. \(2024\)](#). As discussed in the main text, these costs help match



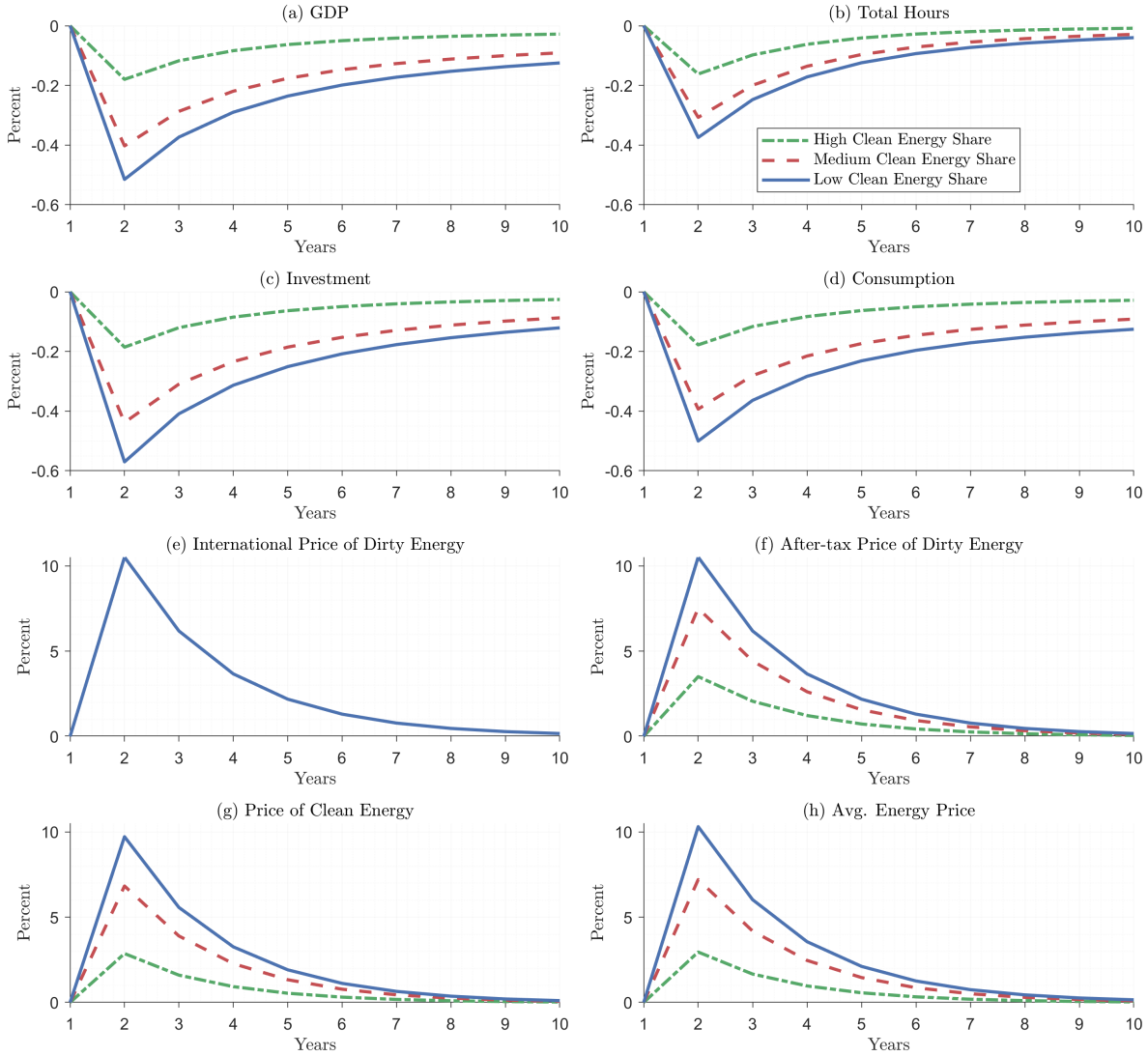
Notes: Change in GDP volatility (standard deviation) in the medium and high clean-energy economies relative to the low clean-energy baseline. The standard deviation is computed from simulations of a long series of shocks to the international price of dirty energy. Robustness check with a lower elasticity of substitution between clean and dirty energy.

Figure C.6: GDP volatility—robustness: Lower elasticity of substitution between clean and dirty energy)

empirical pass-through from international dirty-energy prices to average energy prices and play a key role in the counterfactual exercises in Section 5.3.

To assess robustness, we re-run the analysis without adjustment costs. Figure C.7 reports impulse responses to a 10% increase in the international price of dirty energy (Panel e), and Fig. C.8 reports the corresponding changes in GDP volatility. Without adjustment costs, pass-through from international dirty-energy prices to clean-energy prices increases relative to the baseline. For example, in the low clean-energy economy, the price of clean energy rises nearly one-for-one with the international price (Panel g), as does the average energy price (Panel h). Consequently, the impact of the shock on GDP is slightly larger than in the baseline (compare Panel a with Fig. 1 in the main text).

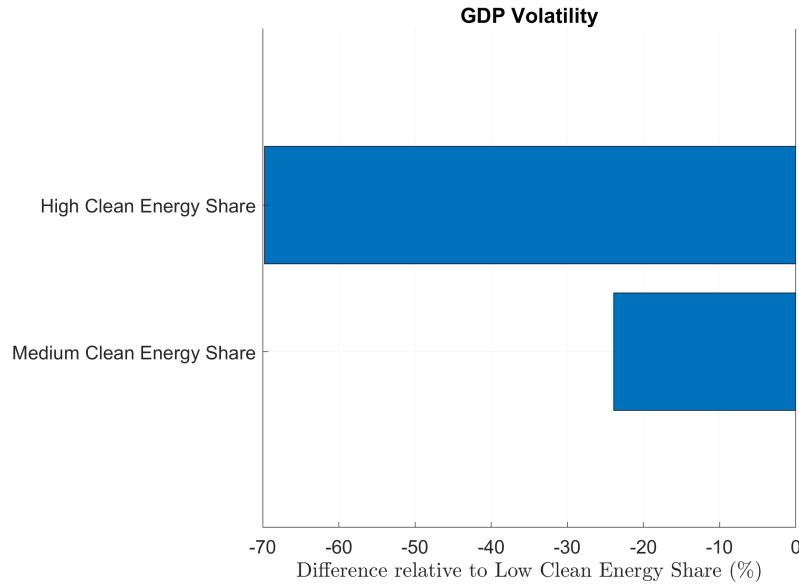
Importantly, the main results remain robust: as the clean-energy share increases, the



Notes: Impulse response functions to a 10% increase in the international price of dirty energy (Panel e). Blue solid line: low clean-energy share economy. Red dashed line: medium clean-energy share economy. Green dotted line: high clean-energy-share economy. Increases in the clean-energy share are achieved through higher carbon taxes. Robustness check without adjustment costs between clean and dirty energy.

Figure C.7: Effects of an increase in the international price of dirty energy—Robustness: No adjustment cost between clean and dirty energy

economy becomes more resilient to fluctuations in international energy prices. The GDP impact in the low clean-energy economy is -0.52% (vs. -0.43% in the baseline), while in the high clean-energy economy it is -0.18% (vs. -0.08%). This pattern is also reflected in GDP volatility (Fig. C.8), where the standard deviation falls by about 70% in the high



Notes: Change in GDP volatility (standard deviation) in the medium and high clean-energy economies relative to the low clean-energy baseline. The standard deviation is computed from simulations of a long series of shocks to the international price of dirty energy. Robustness check without adjustment costs between clean and dirty energy.

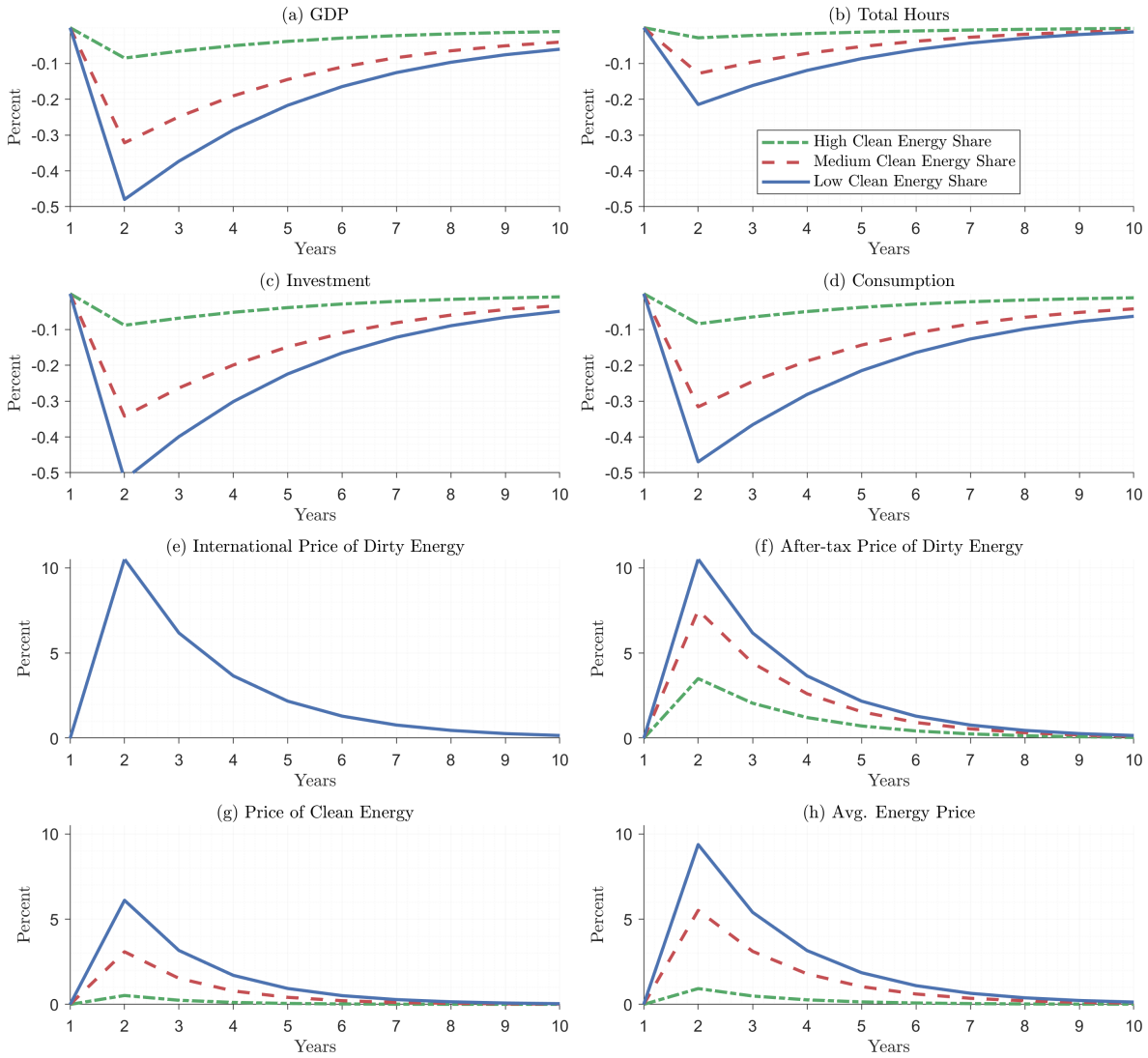
Figure C.8: GDP volatility—Robustness: No adjustment cost between clean and dirty energy

clean-energy economy relative to the low clean-energy case, broadly consistent with the baseline.

Innovation The calibration of innovation parameters in the baseline relies on Spanish firm-level data, following [Atkeson and Burstein \(2019\)](#) and [Domínguez-Díaz et al. \(2025\)](#). A key parameter is ϕ in (19) and (17), which governs intertemporal knowledge spillovers in research and captures the notion that “ideas are getting harder to find” ([Bloom et al., 2020](#)). In the baseline, $\phi = -1.6$. Given uncertainty around this parameter, we re-run the analysis with $\phi = -1.01$, which effectively eliminates intertemporal knowledge spillovers.

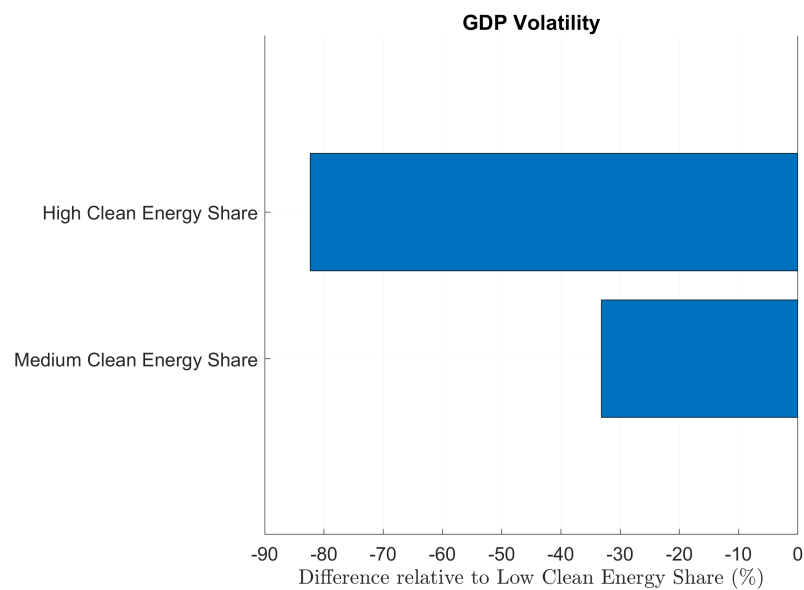
Figure C.9 reports the responses to a 10% increase in the international price of dirty

energy (Panel e) across the three clean-energy scenarios under this alternative calibration. The overall impact is very similar to the baseline. Crucially, the main result remains: higher clean-energy shares reduce vulnerability to external energy price shocks. This is evident both in the impulse responses (Panel a of Fig. [C.9](#)) and in the corresponding reductions in GDP volatility shown in Fig. [C.10](#).



Notes: Impulse response functions to a 10% increase in the international price of dirty energy (Panel e). Blue solid line: low clean-energy share economy. Red dashed line: medium clean-energy share economy. Green dotted line: high clean-energy share economy. Increases in the clean-energy share are achieved through higher carbon taxes. Robustness check with alternative innovation parameters.

Figure C.9: Effects of an increase in the international price of dirty energy—Robustness: Innovation



Notes: Change in GDP volatility (standard deviation) in the medium and high clean-energy economies relative to the low clean-energy baseline. The standard deviation is computed from simulations of a long series of shocks to the international price of dirty energy. Robustness check with alternative innovation parameters.

Figure C.10: GDP volatility—Robustness: Innovation